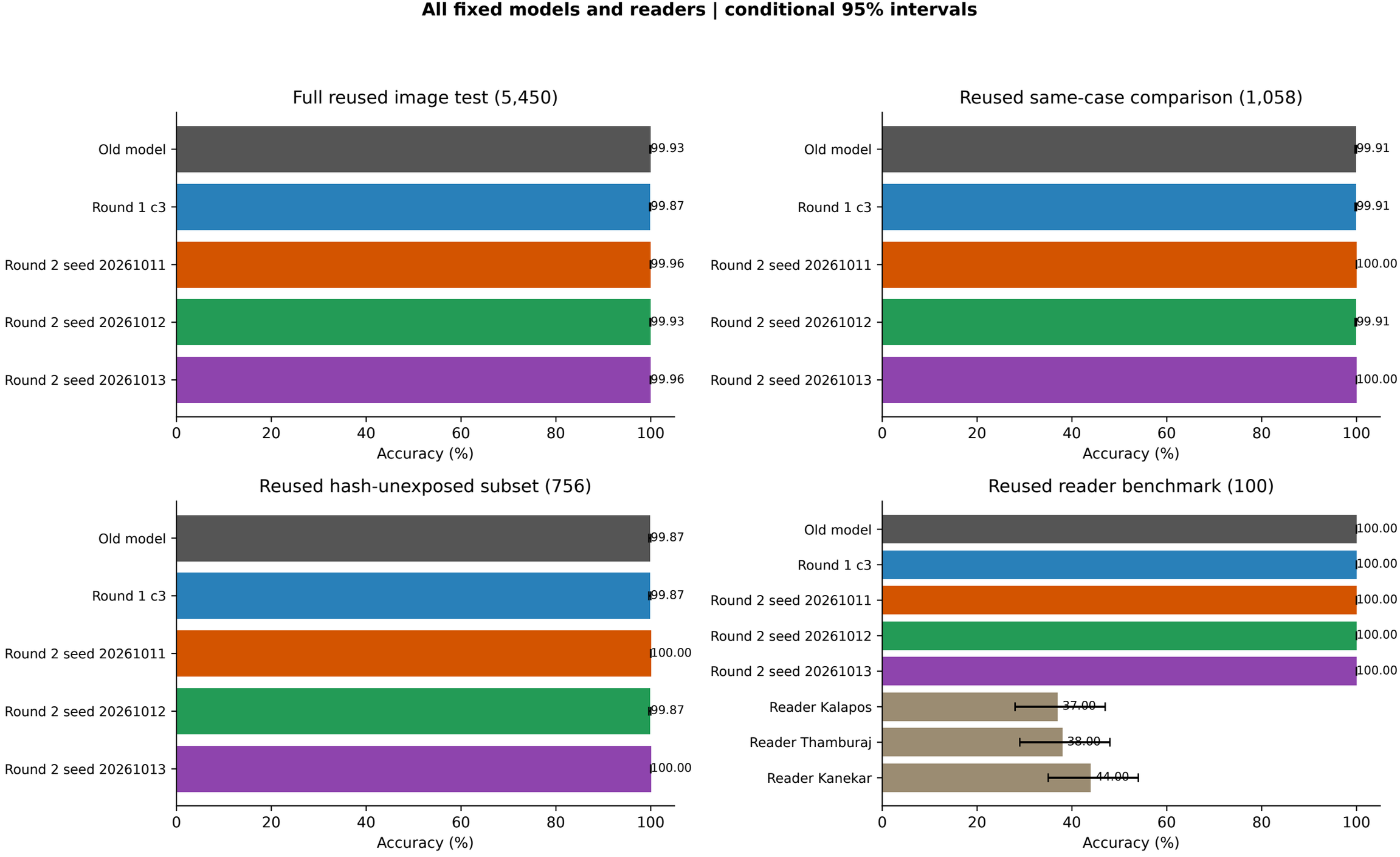
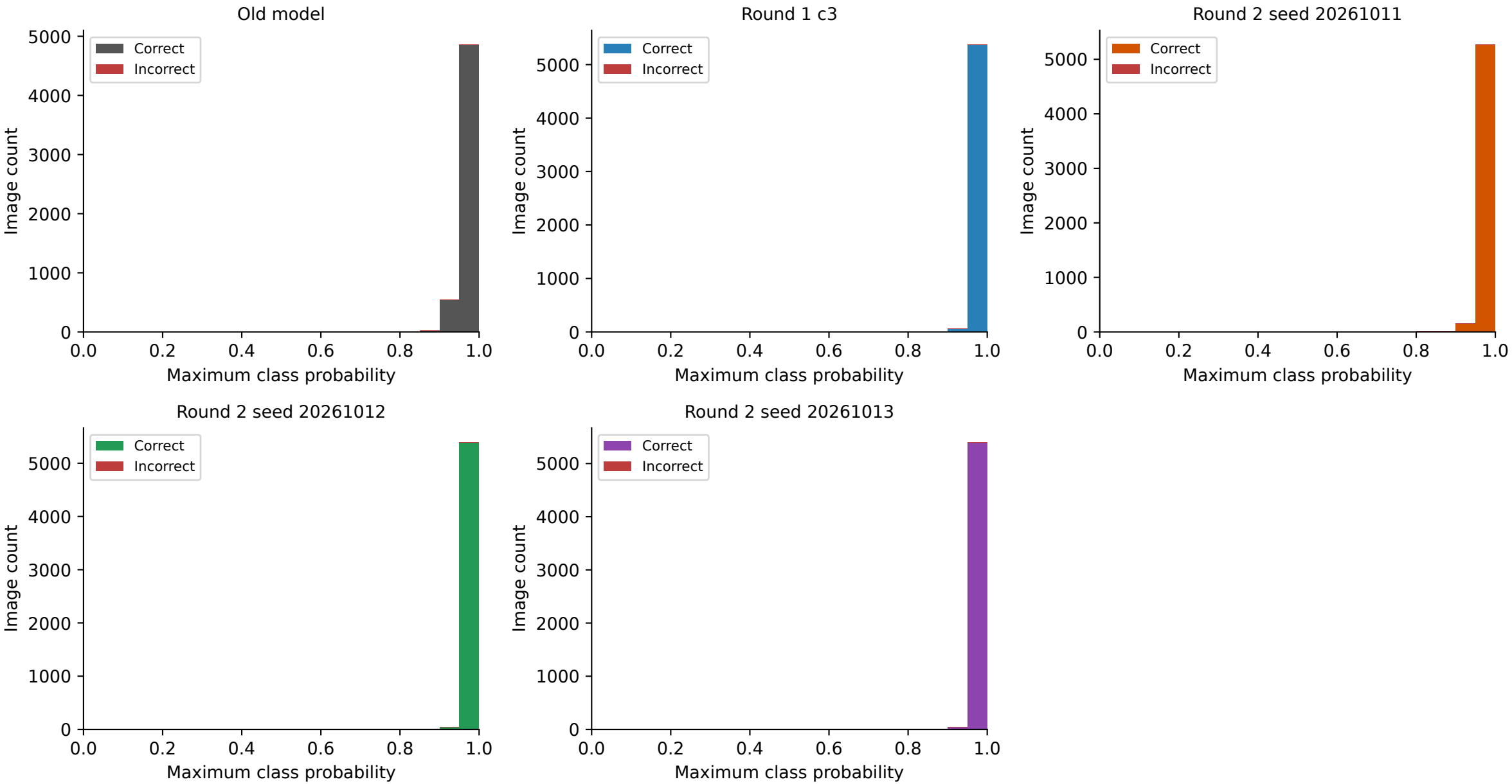


Paired 2,000-draw hash-group bootstrap; original100 image units. Degenerate intervals do not prove certainty.
Three-reader range is descriptive across fixed readers; repeated benchmarks cannot establish independent superiority.

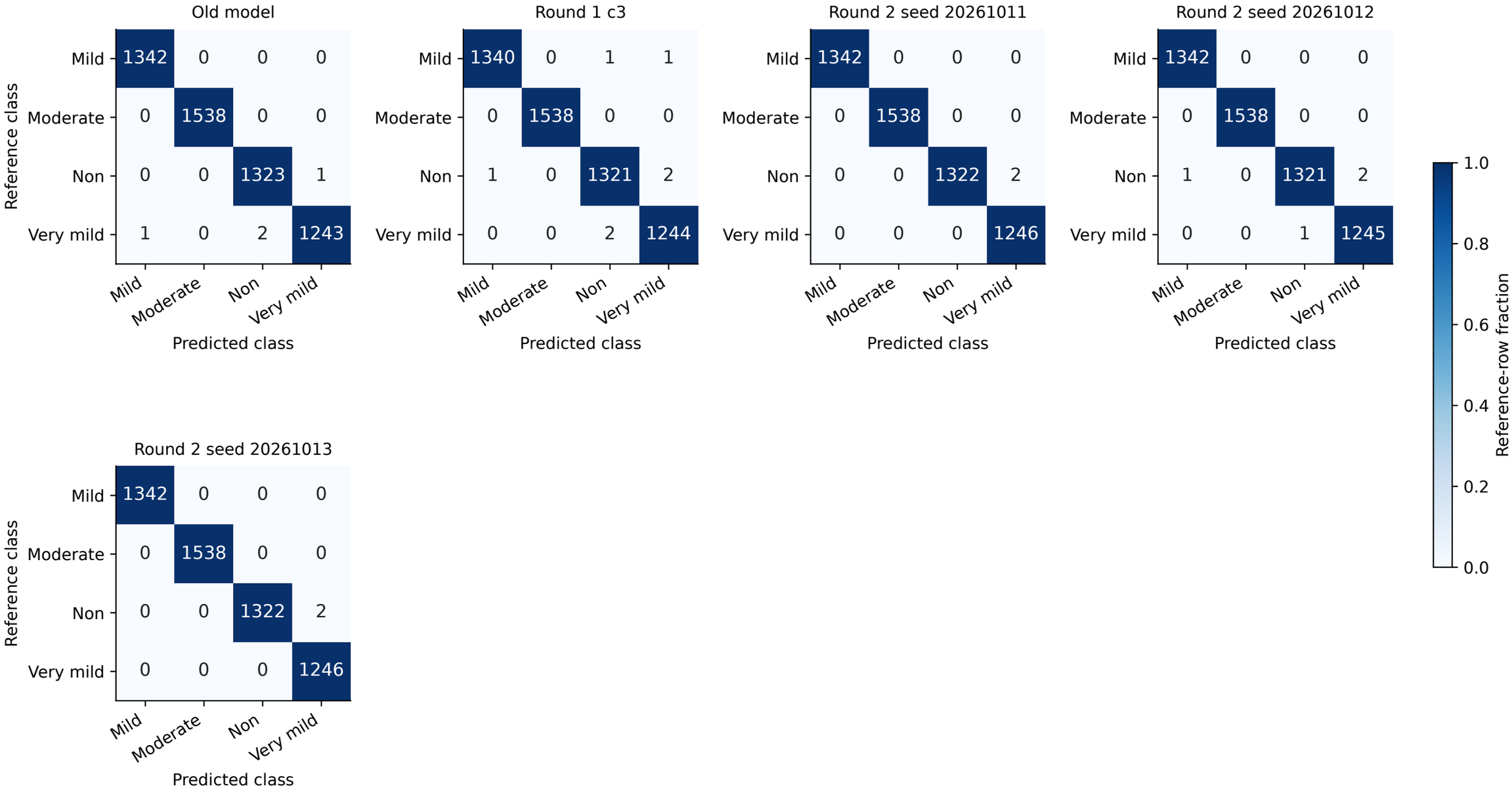
Full reused image test (5,450) | Confidence



Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

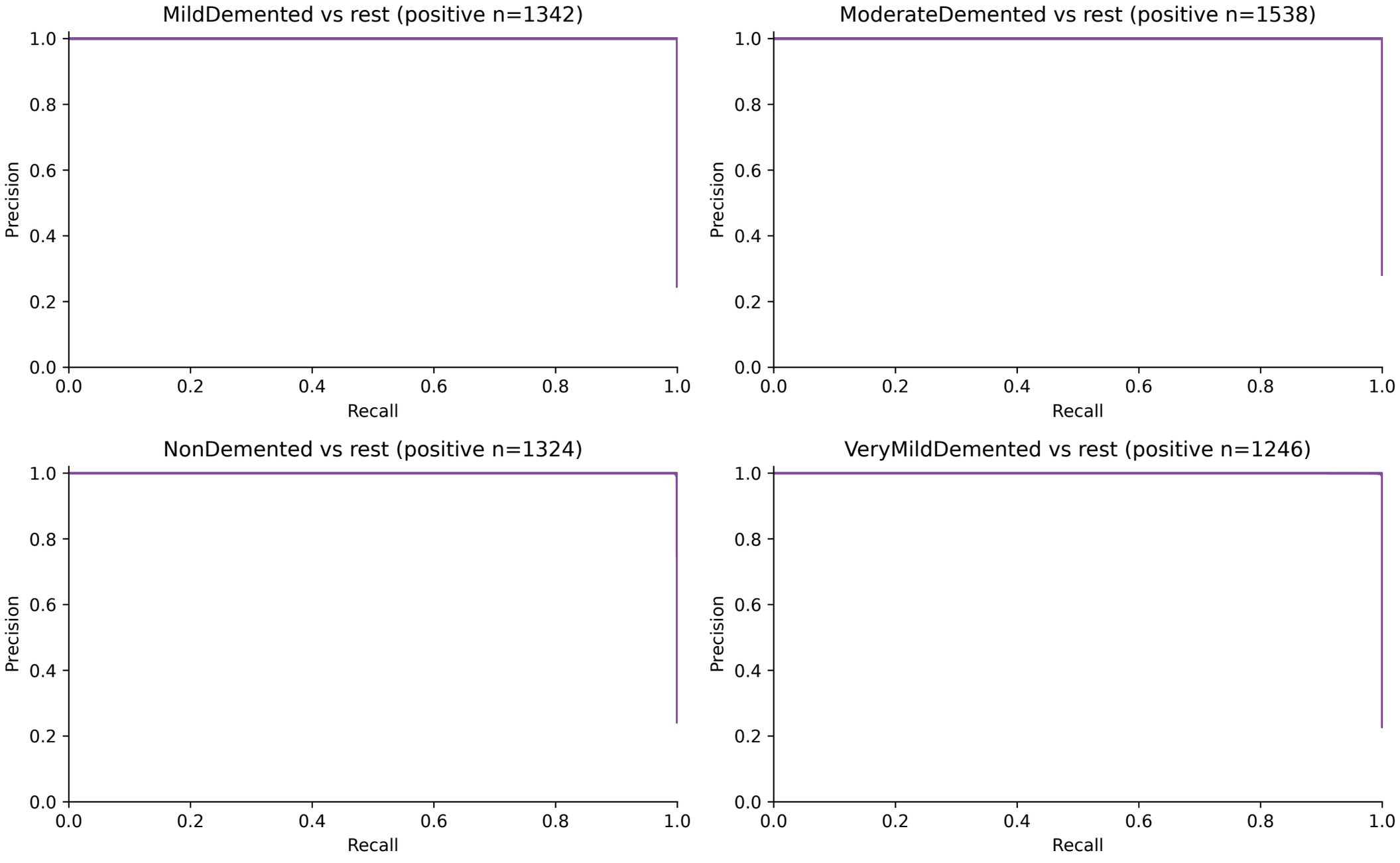
Full reused image test (5,450) | confusion | n=5450

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

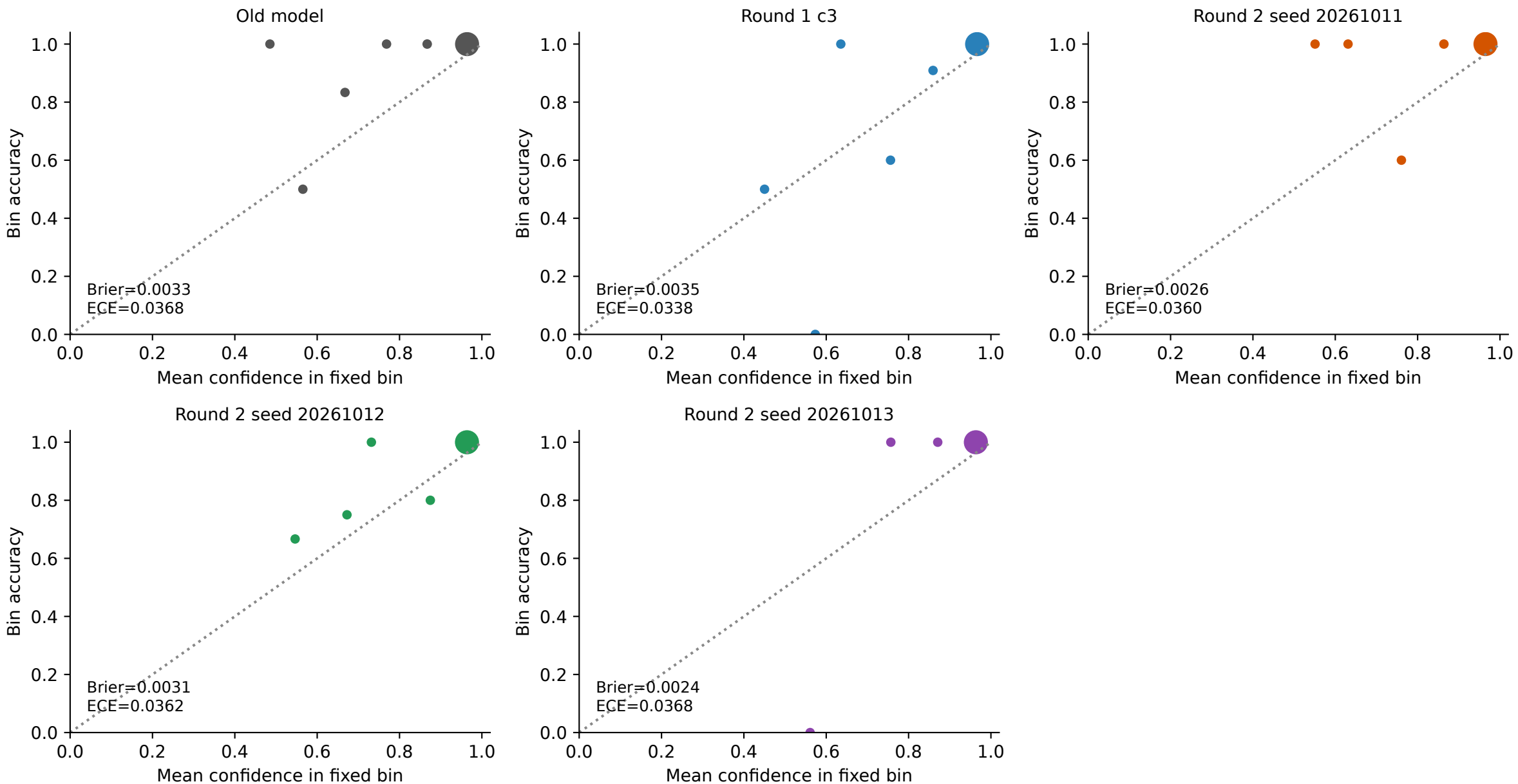


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

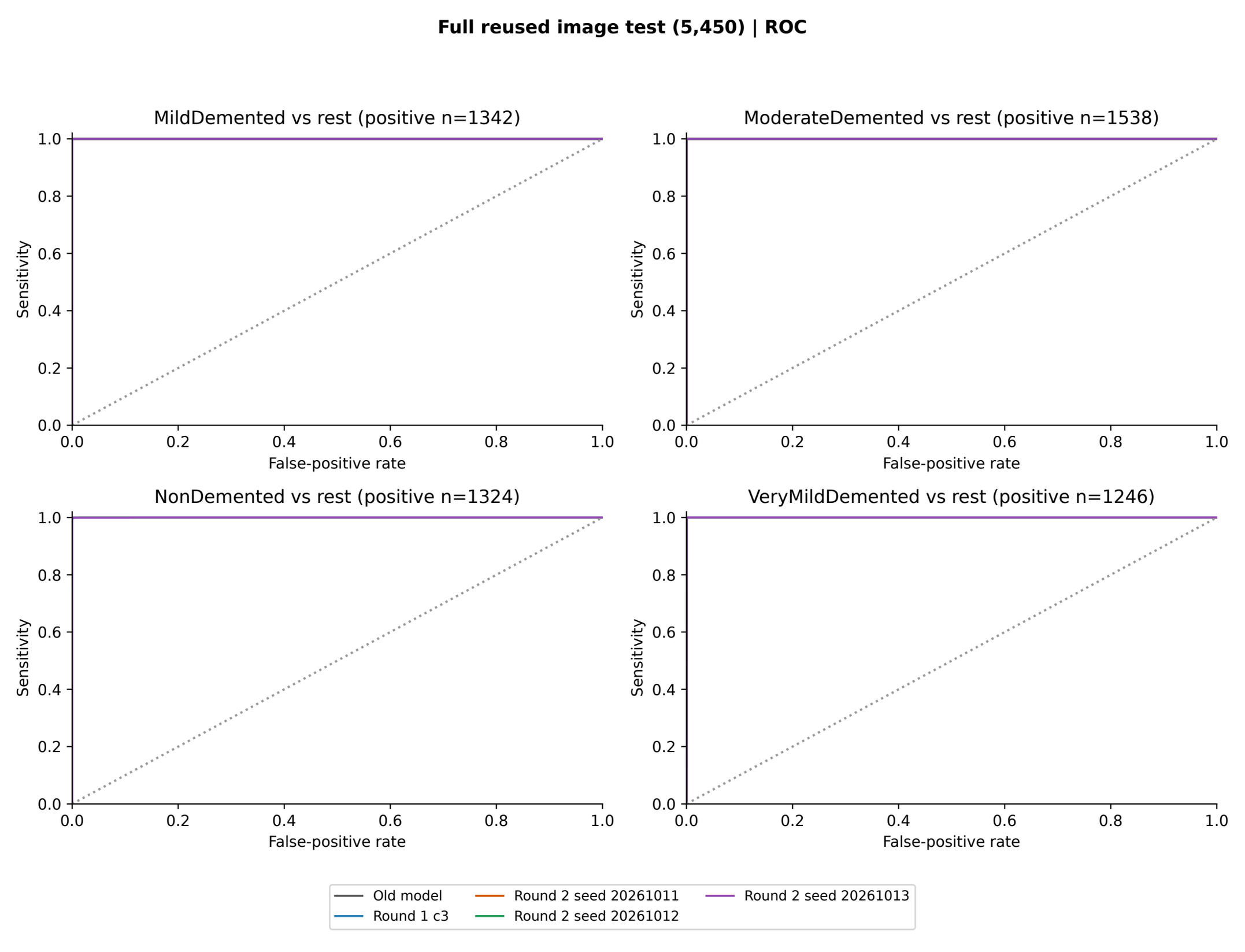
Full reused image test (5,450) | Precision–recall

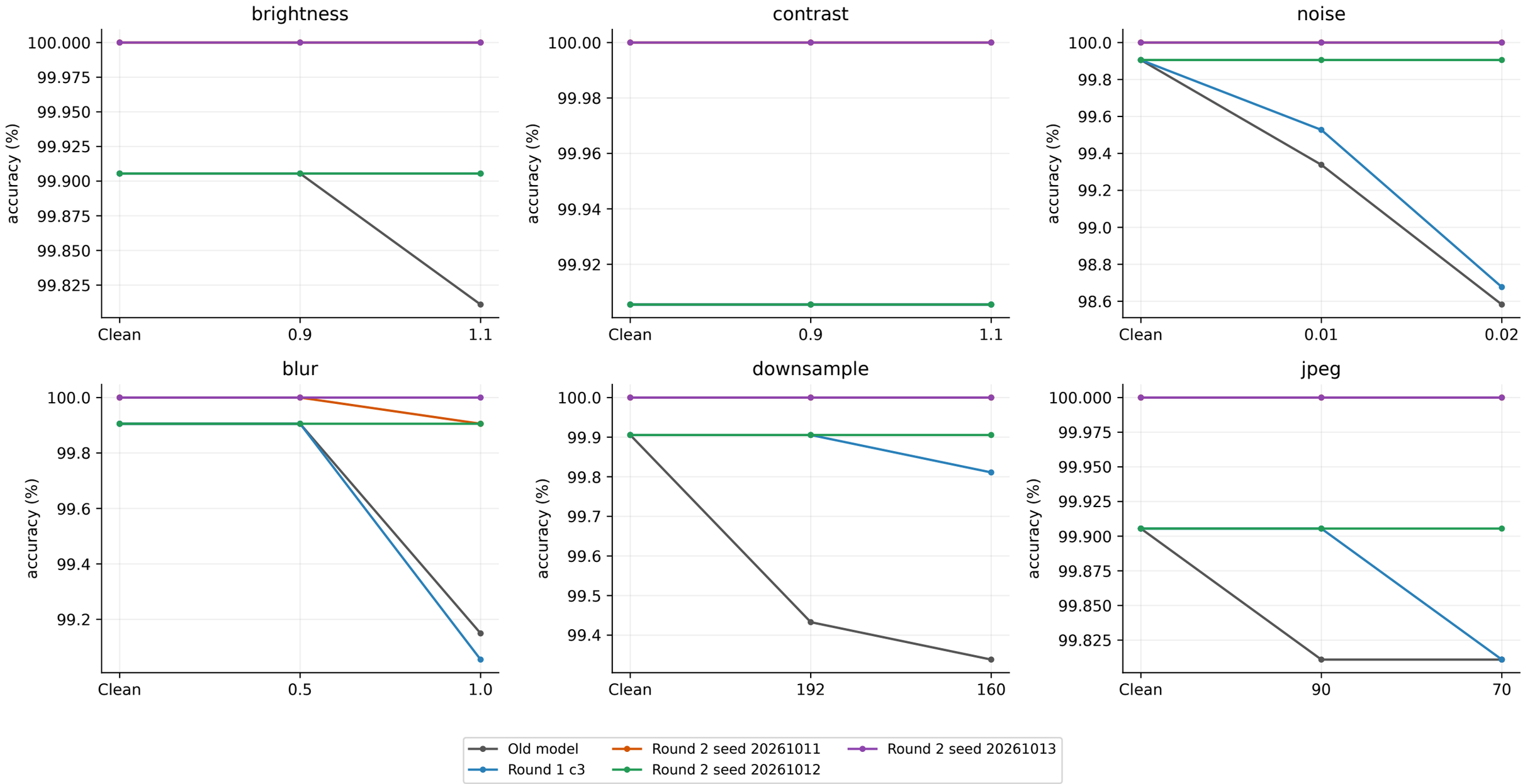


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

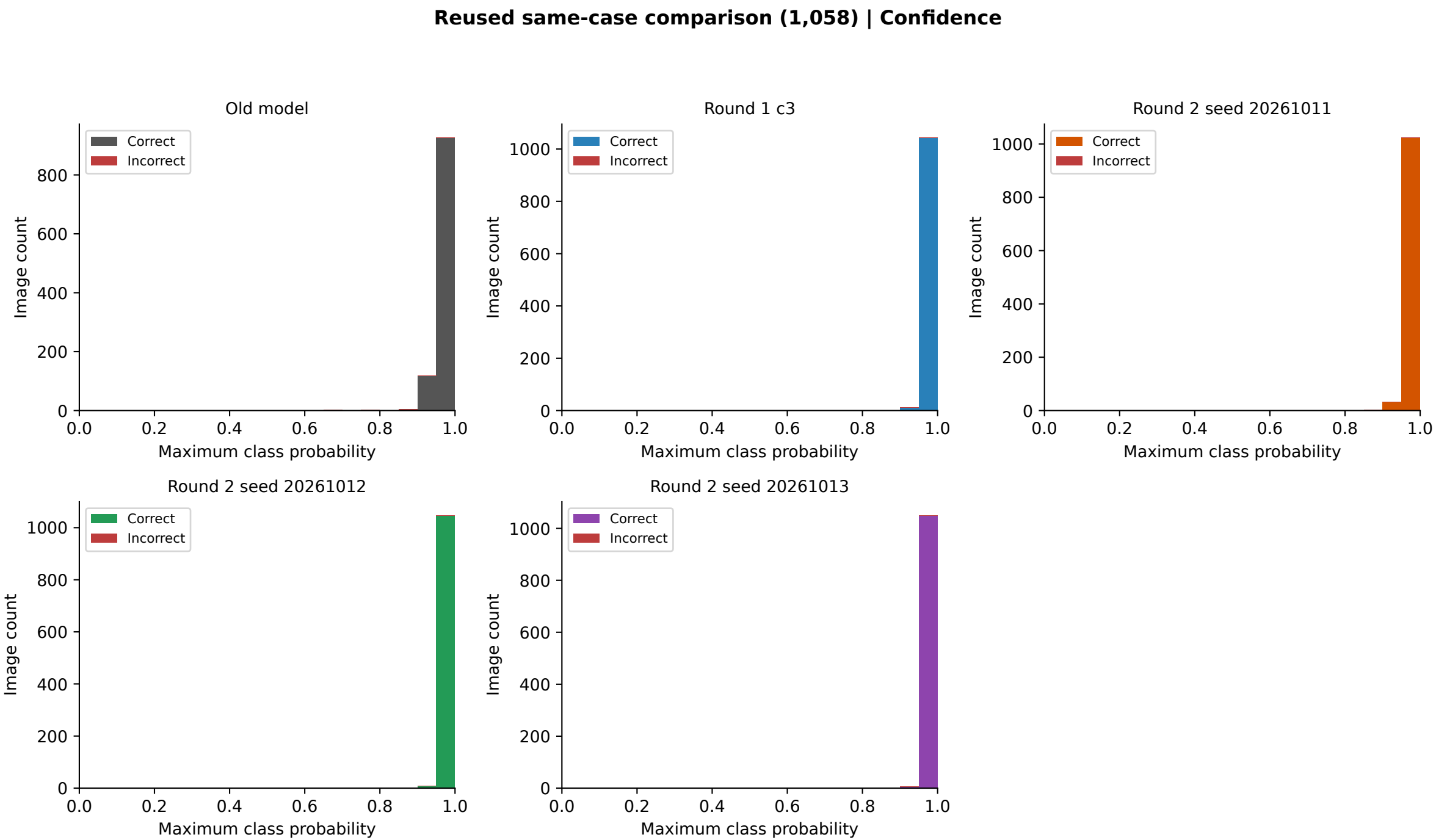


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Full reused image test (5,450) | ROC
MildDemented vs rest (positive n=1342)
ModerateDemented vs rest (positive n=1538)
NonDemented vs rest (positive n=1324)
VeryMildDemented vs rest (positive n=1246)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



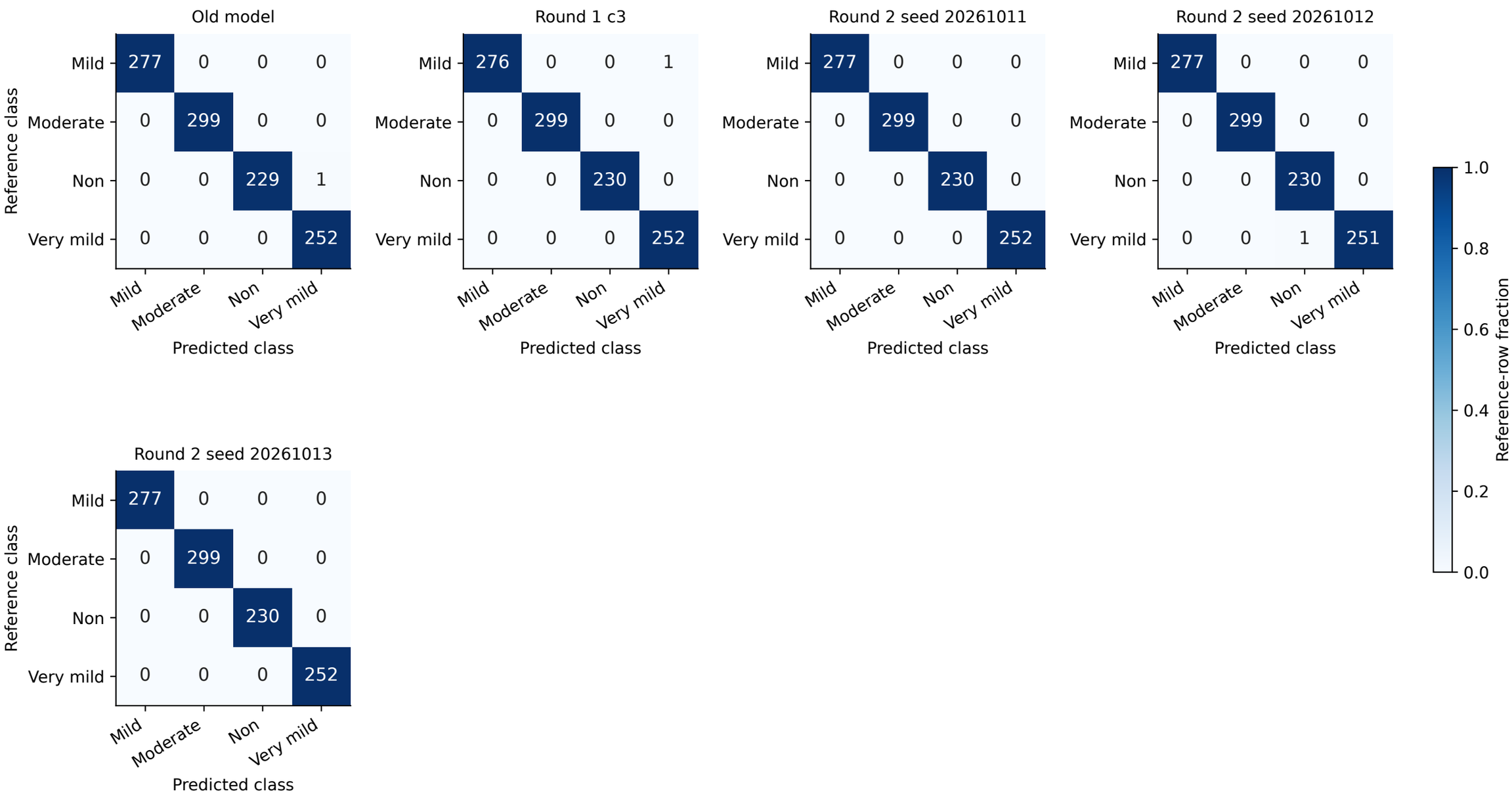
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.



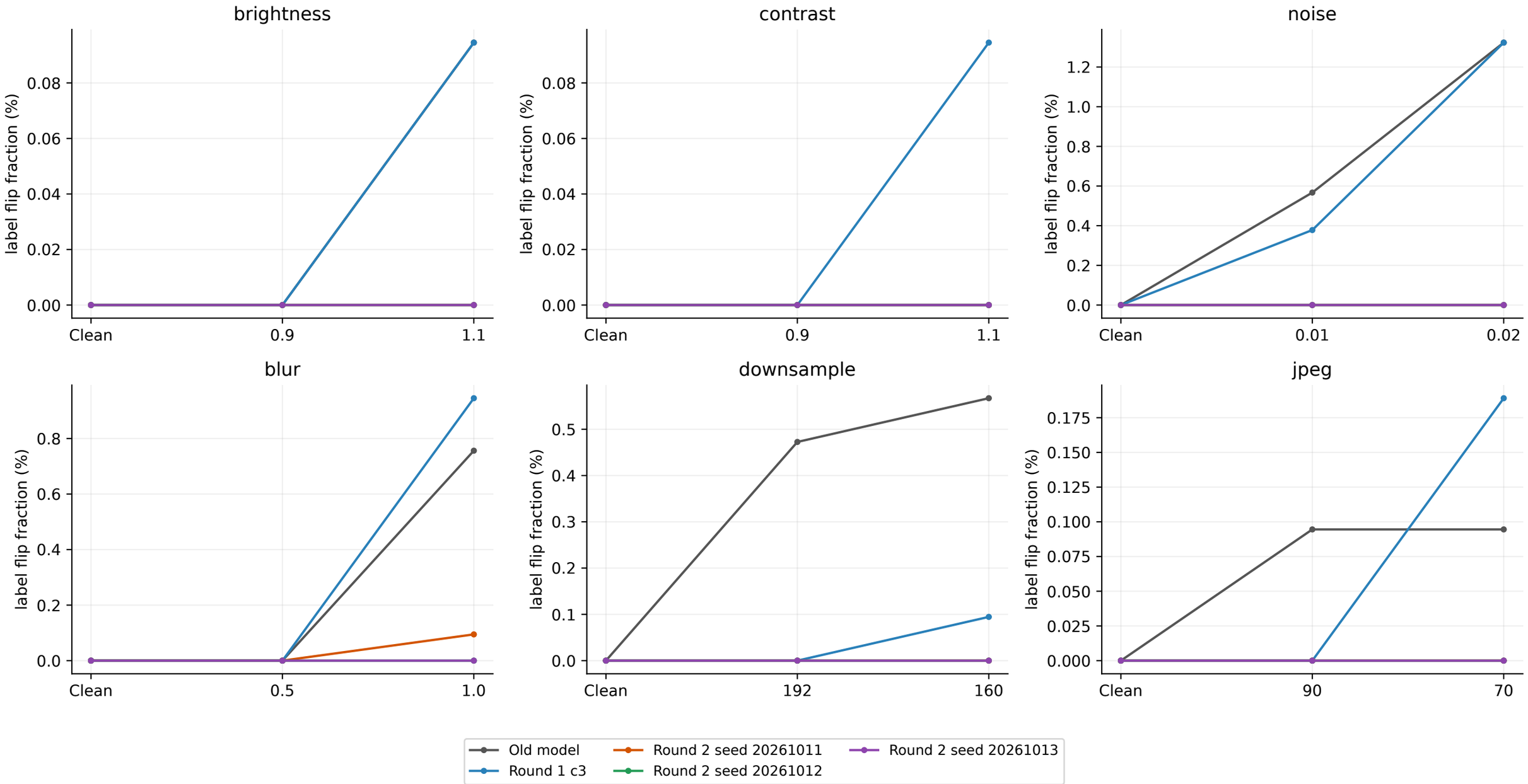
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

Reused same-case comparison (1,058) | confusion | n=1058

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

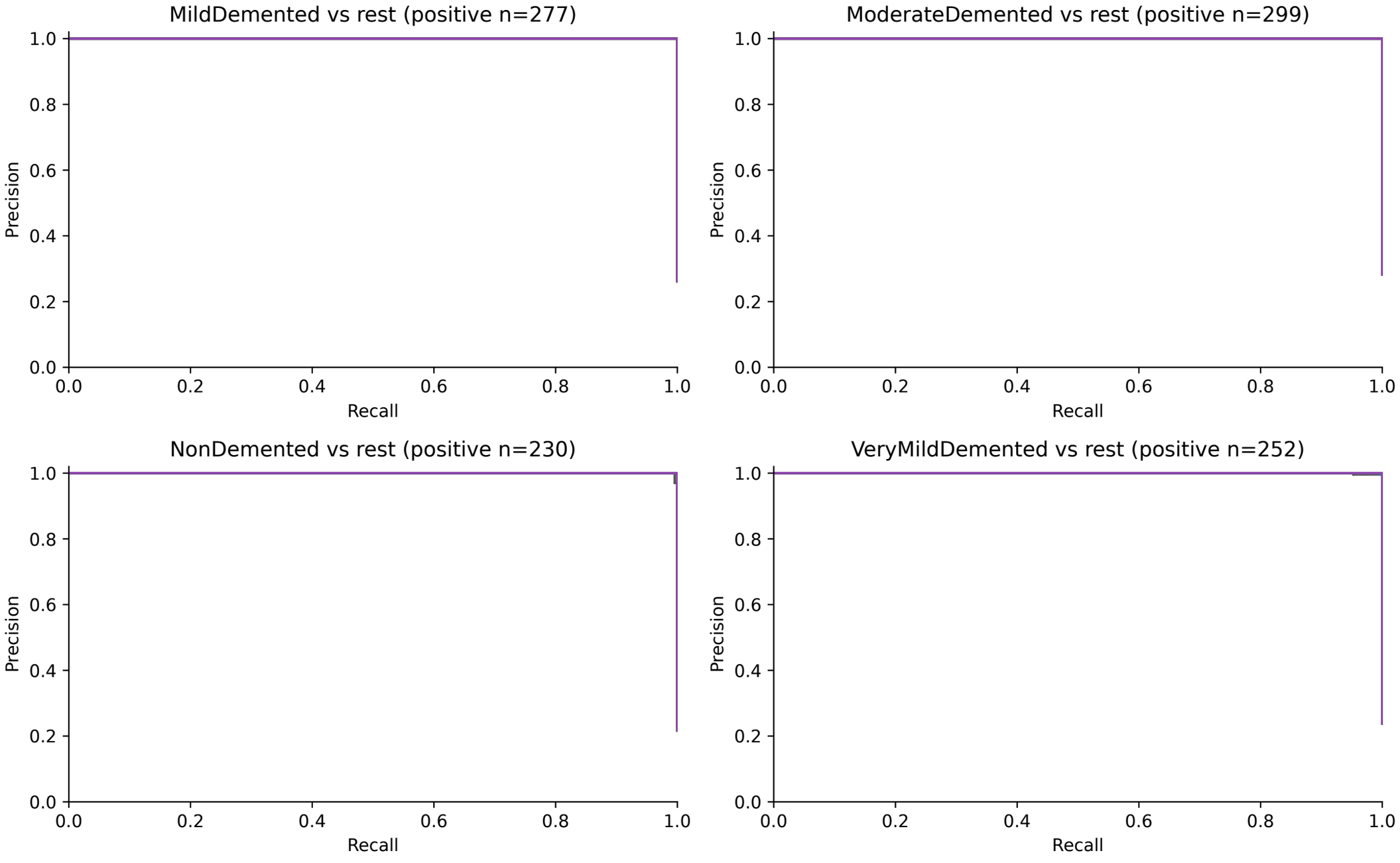


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

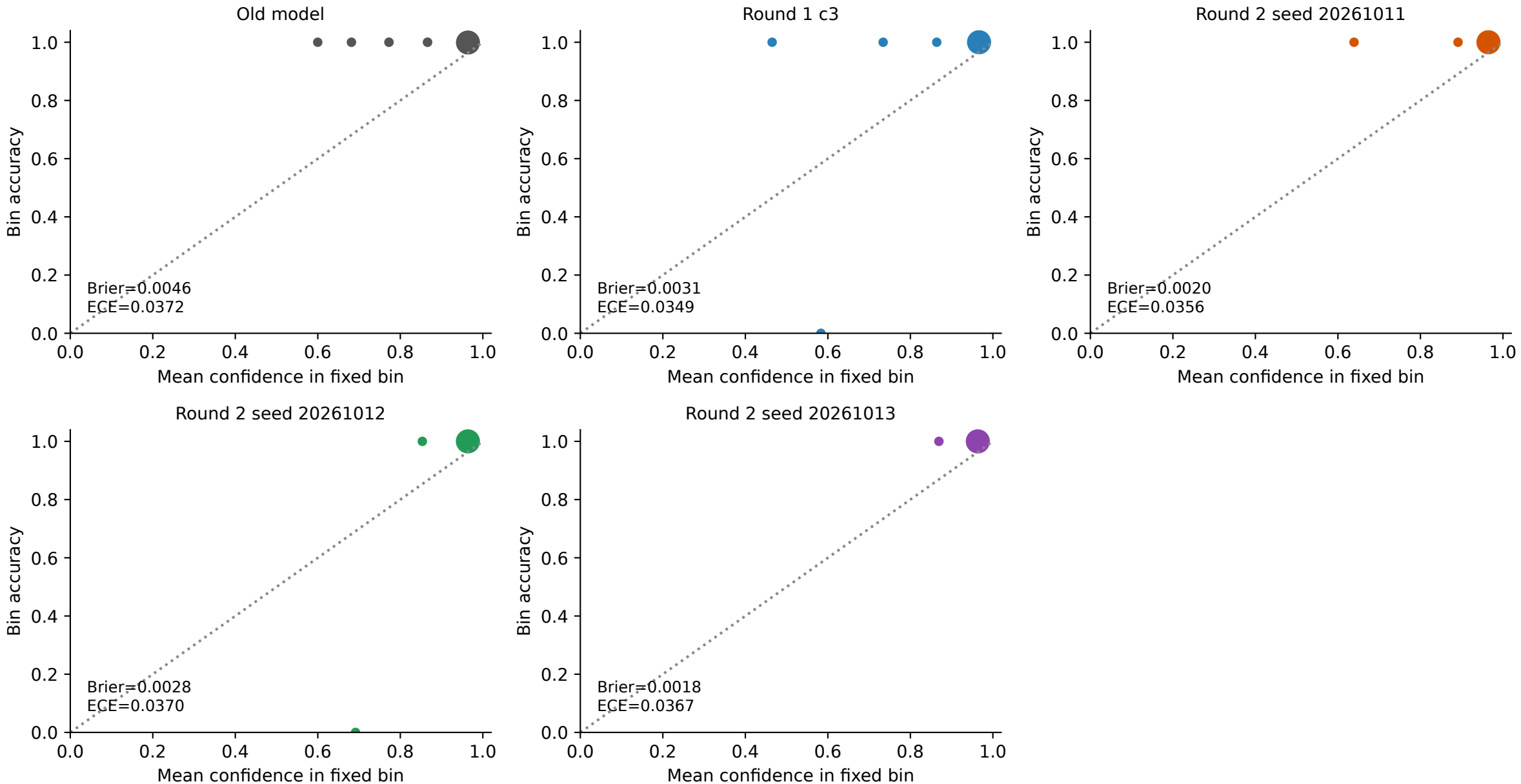


Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

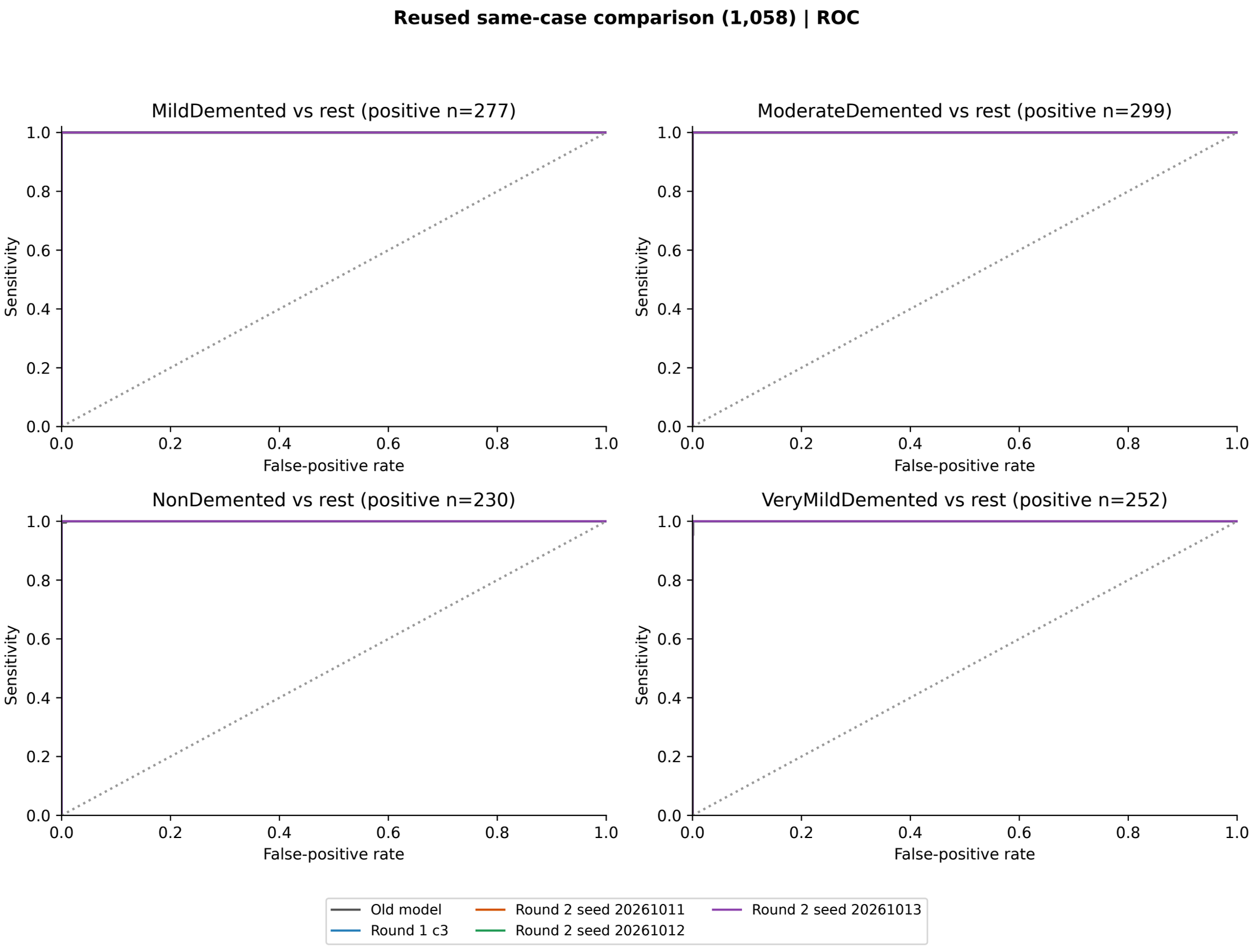
Reused same-case comparison (1,058) | Precision–recall



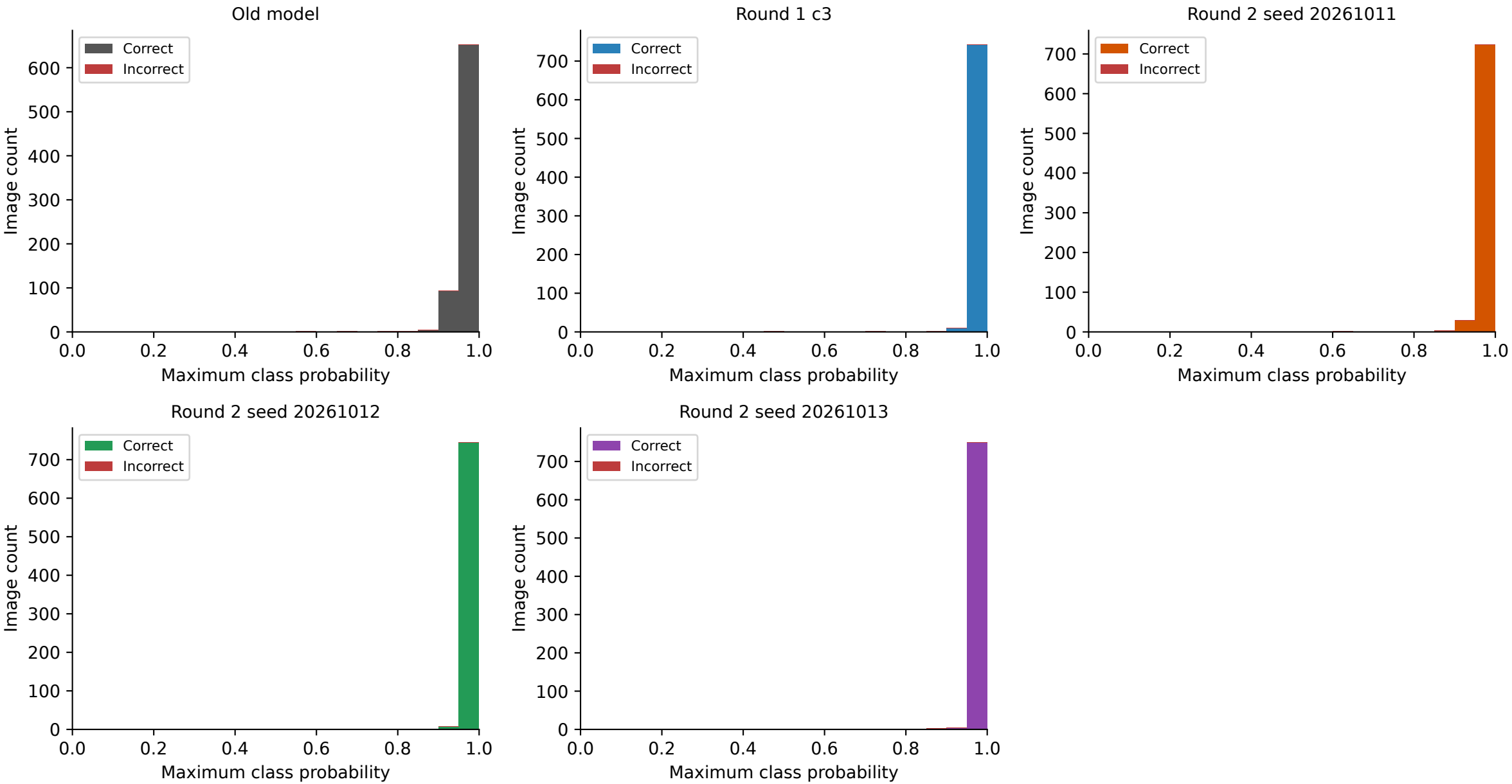
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused same-case comparison (1,058) | ROC
MildDemented vs rest (positive n=277)
ModerateDemented vs rest (positive n=299)
NonDemented vs rest (positive n=230)
VeryMildDemented vs rest (positive n=252)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

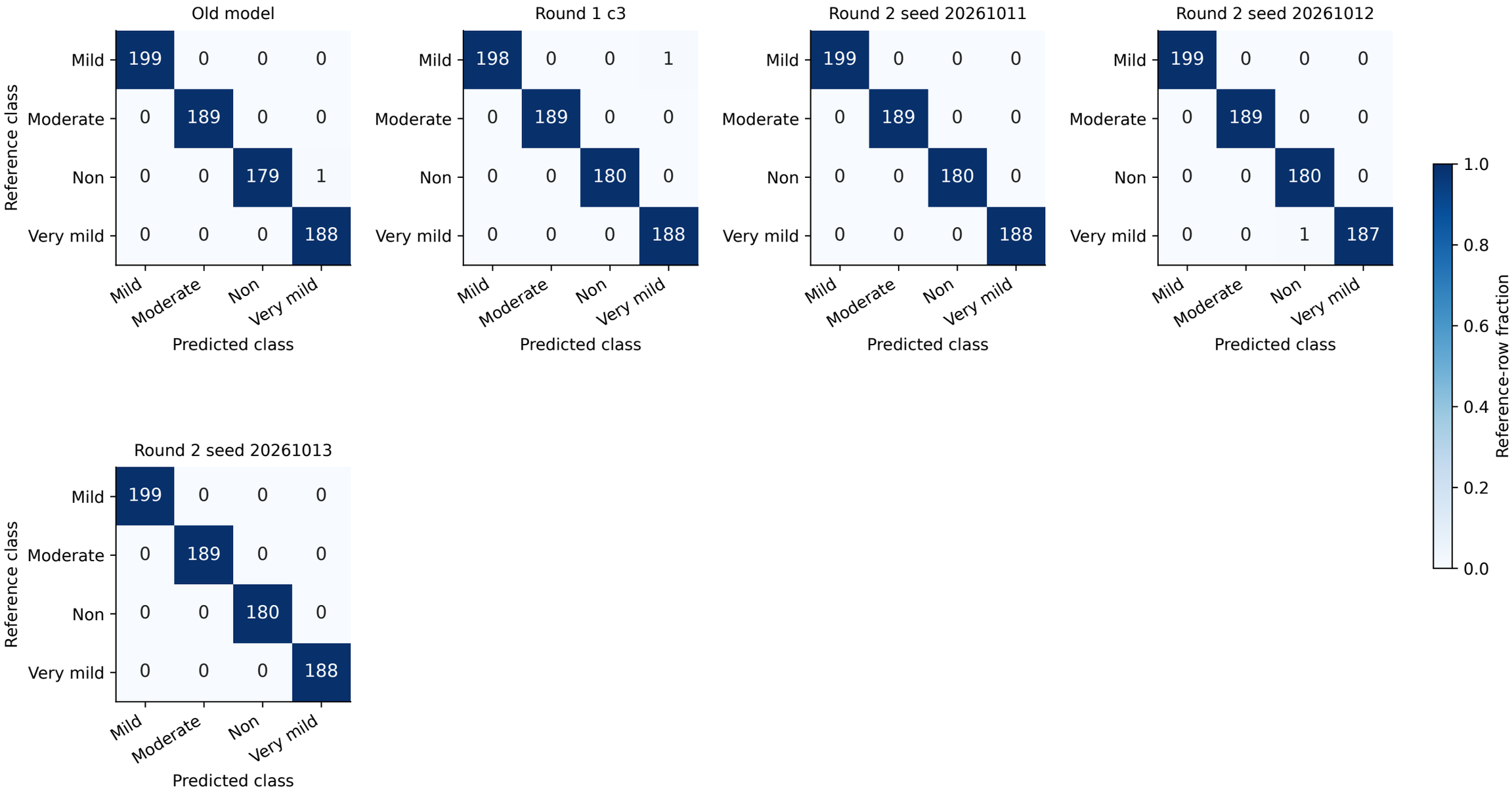
Reused hash-unexposed subset (756) | Confidence



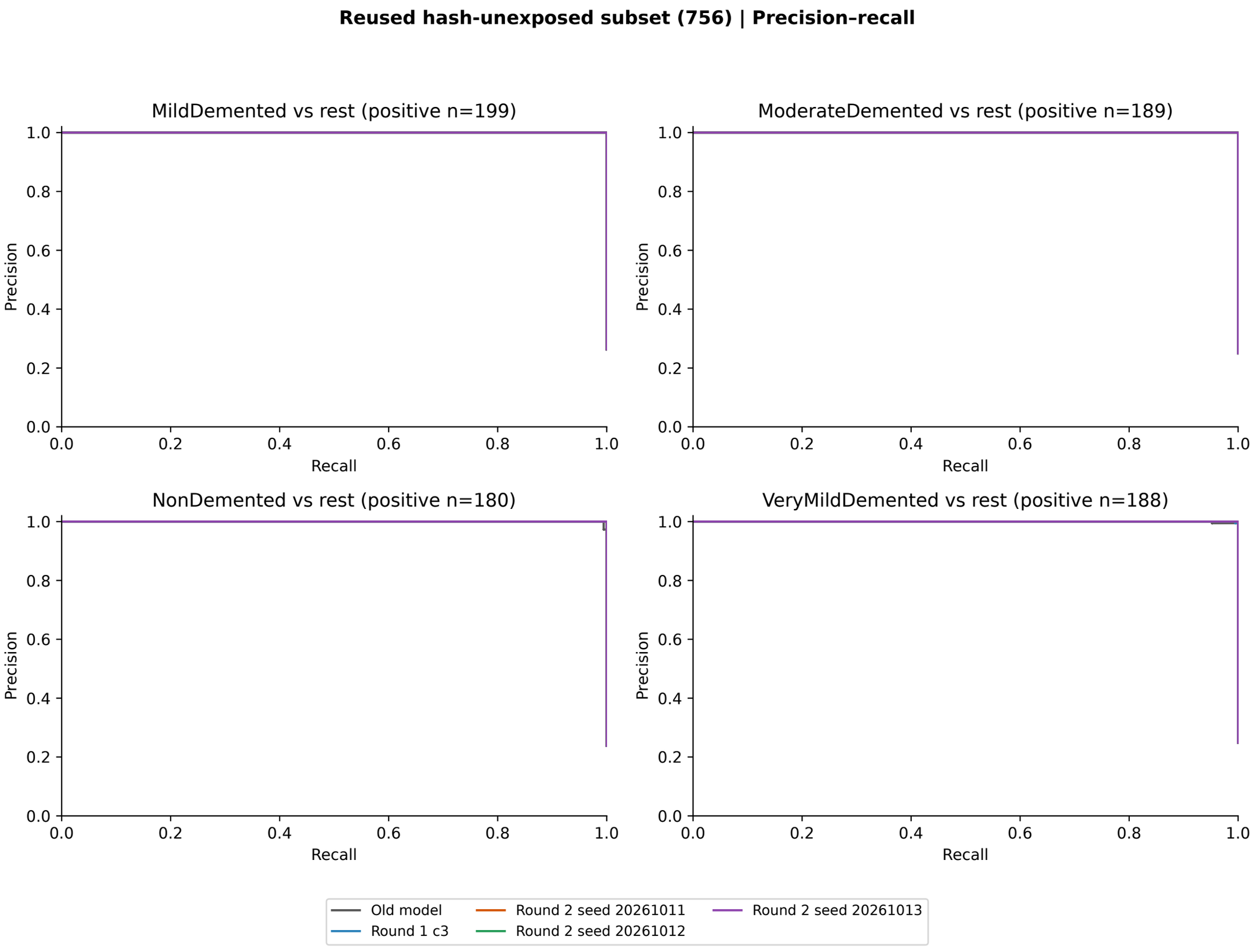
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

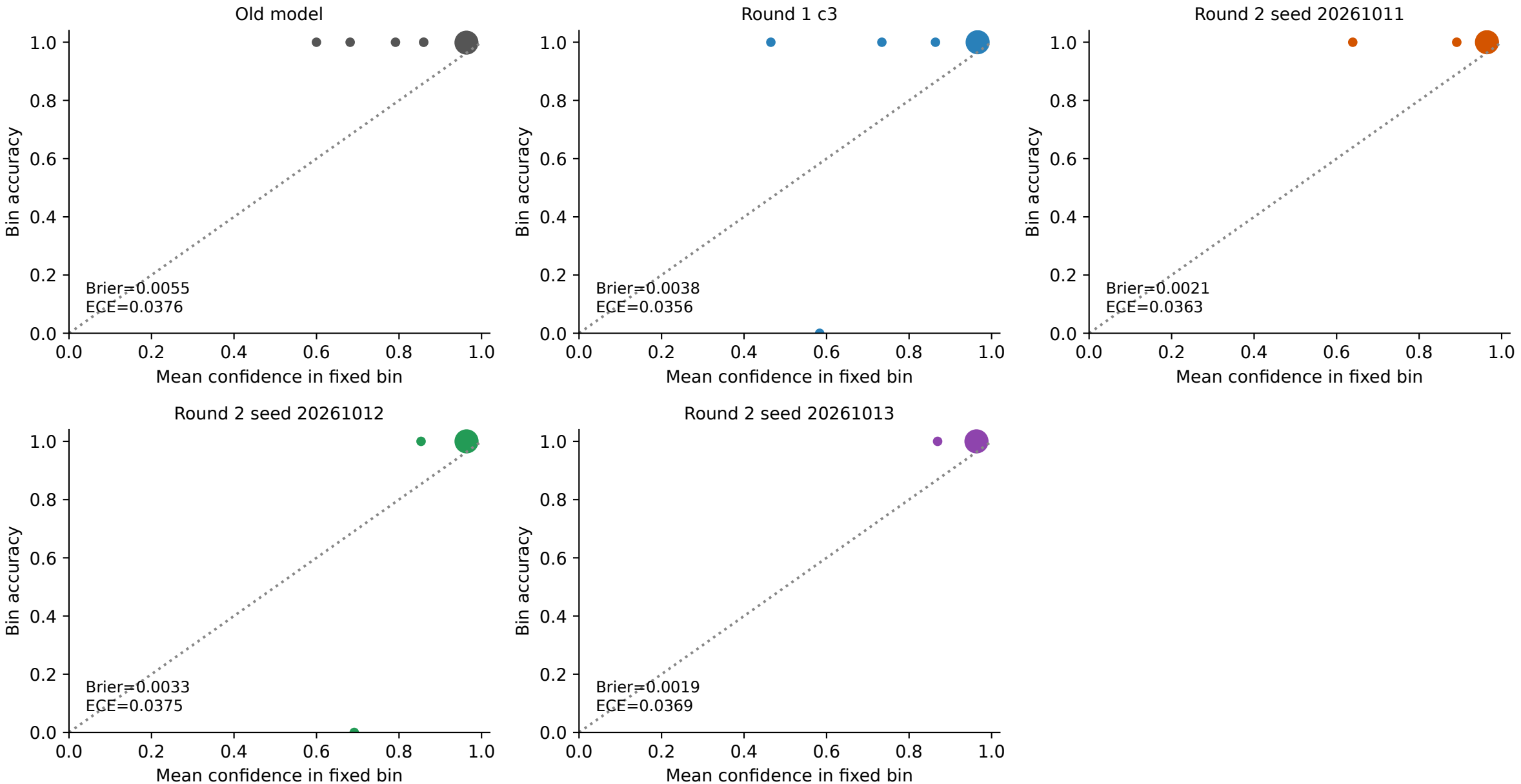
Reused hash-unexposed subset (756) | confusion | n=756

Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.

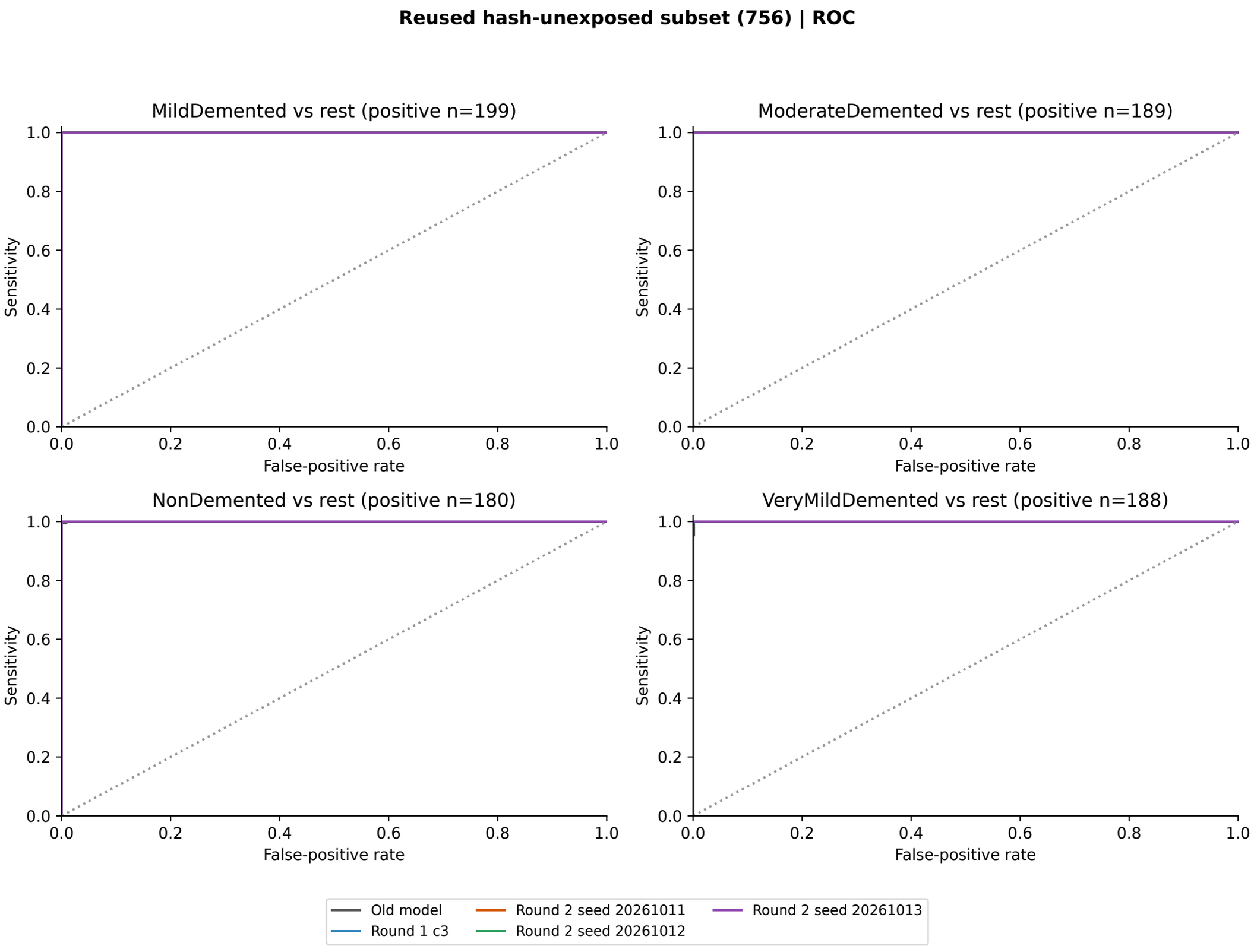


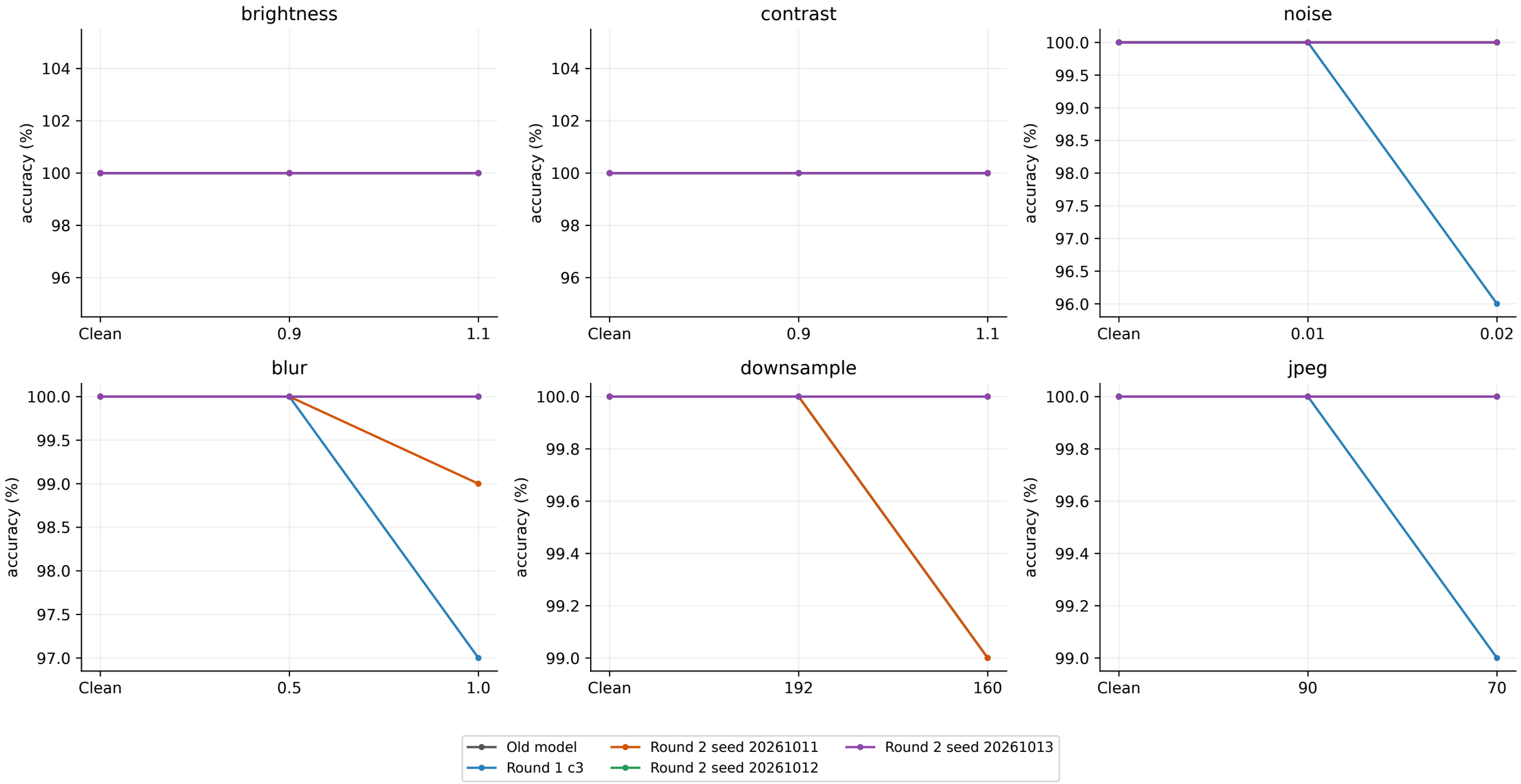
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused hash-unexposed subset (756) | Precision–recall
MildDemented vs rest (positive n=199)
ModerateDemented vs rest (positive n=189)
NonDemented vs rest (positive n=180)
VeryMildDemented vs rest (positive n=188)
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.



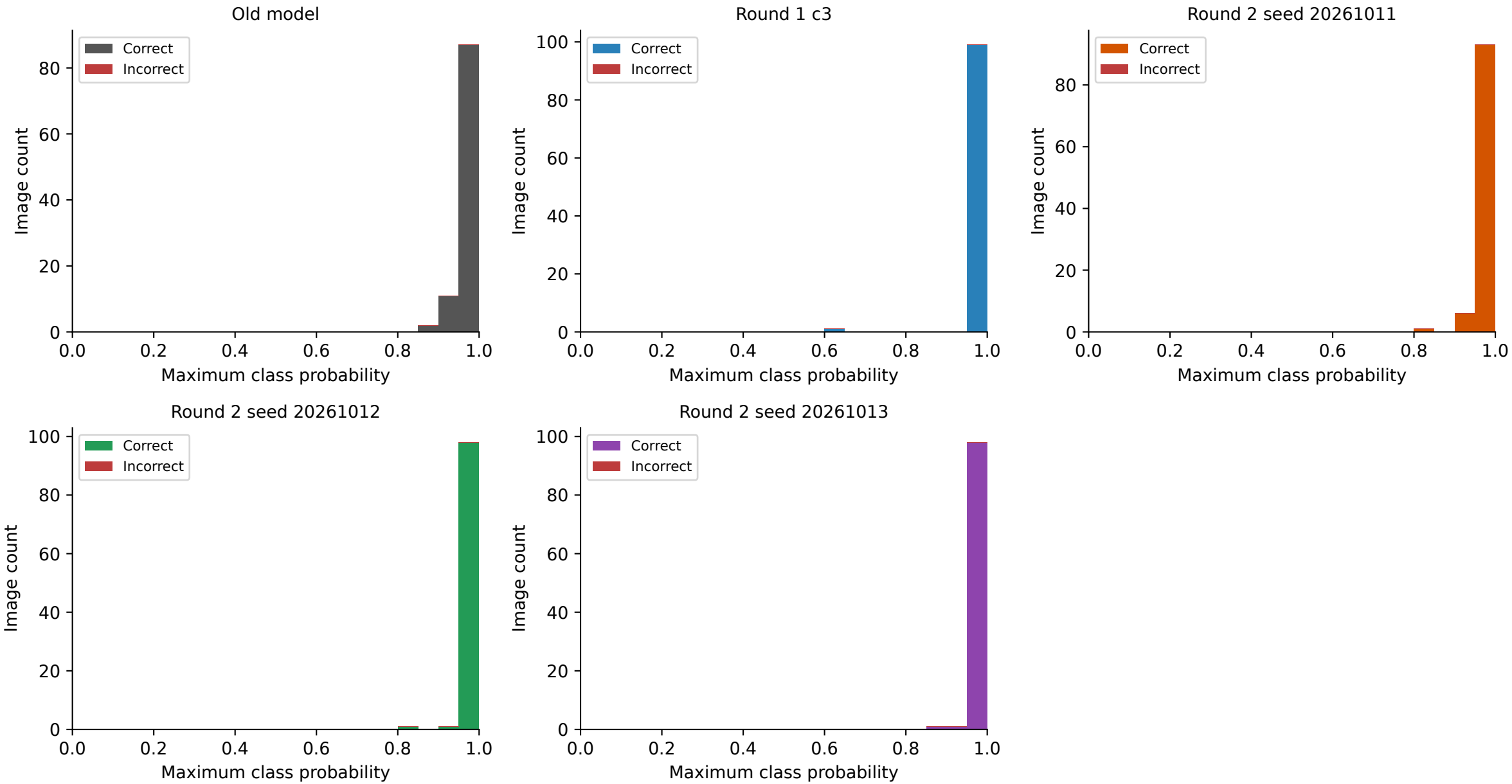
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused hash-unexposed subset (756) | ROC
MildDemented vs rest (positive n=199)
ModerateDemented vs rest (positive n=189)
NonDemented vs rest (positive n=180)
VeryMildDemented vs rest (positive n=188)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

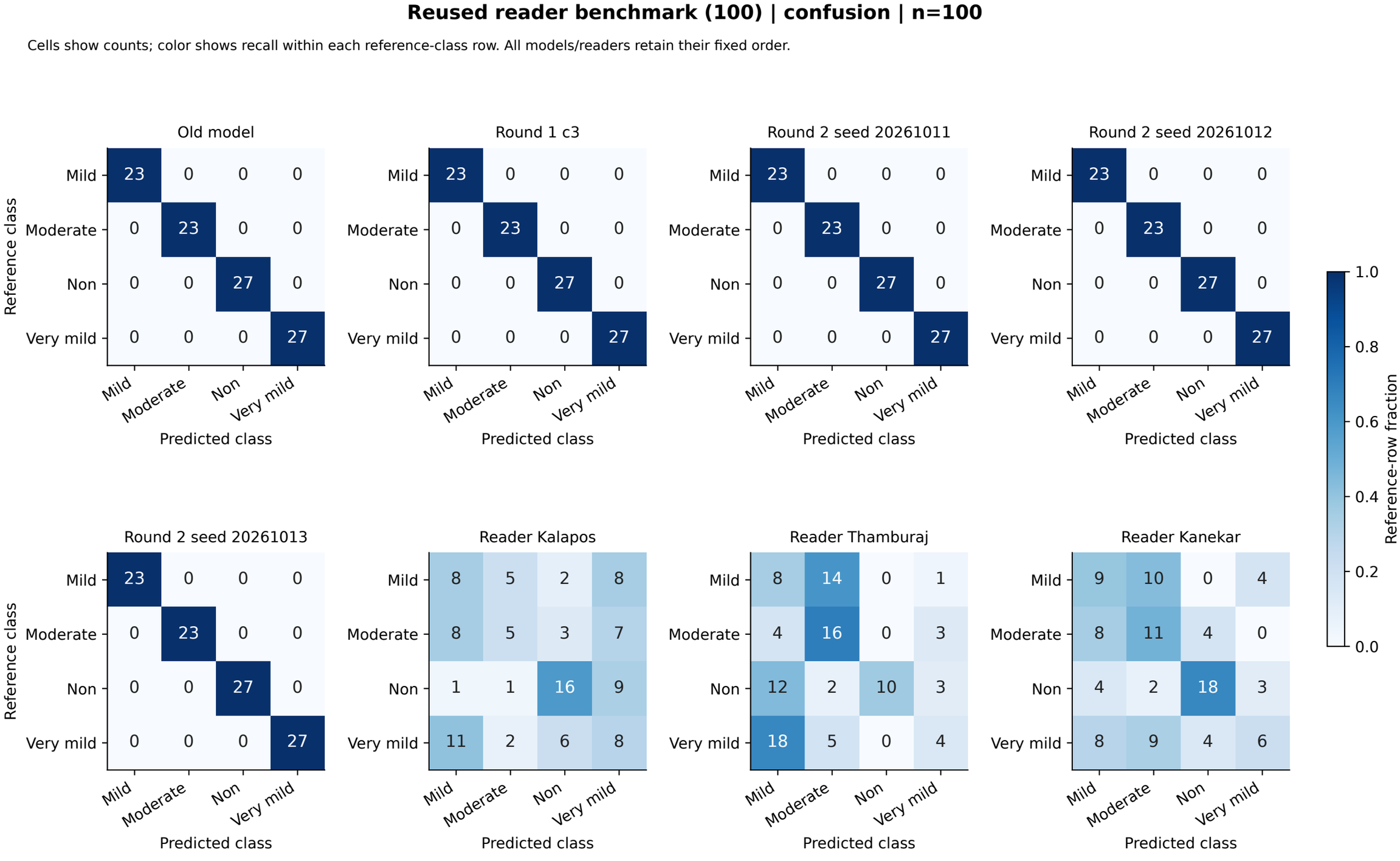


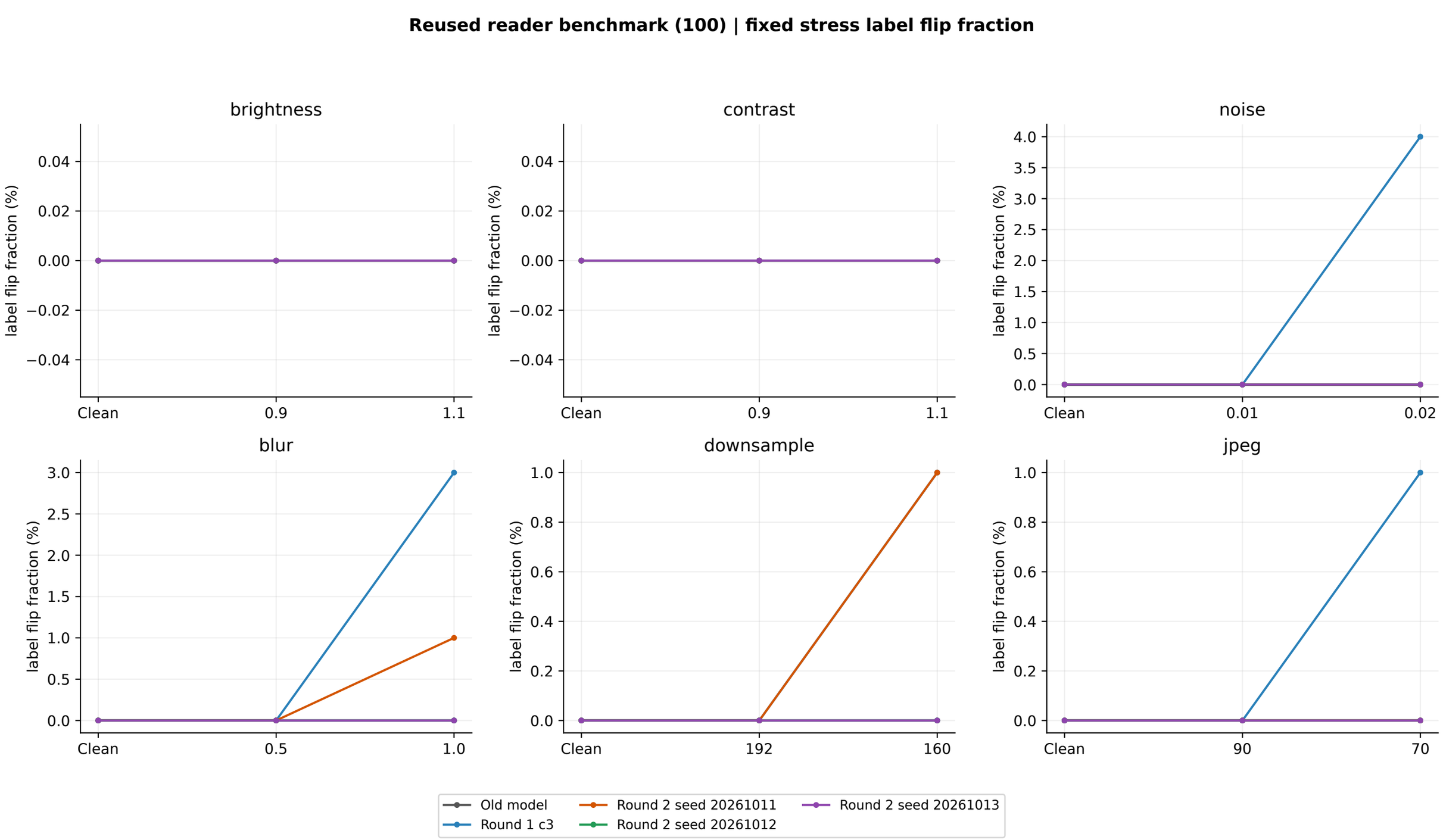
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

Reused reader benchmark (100) | Confidence

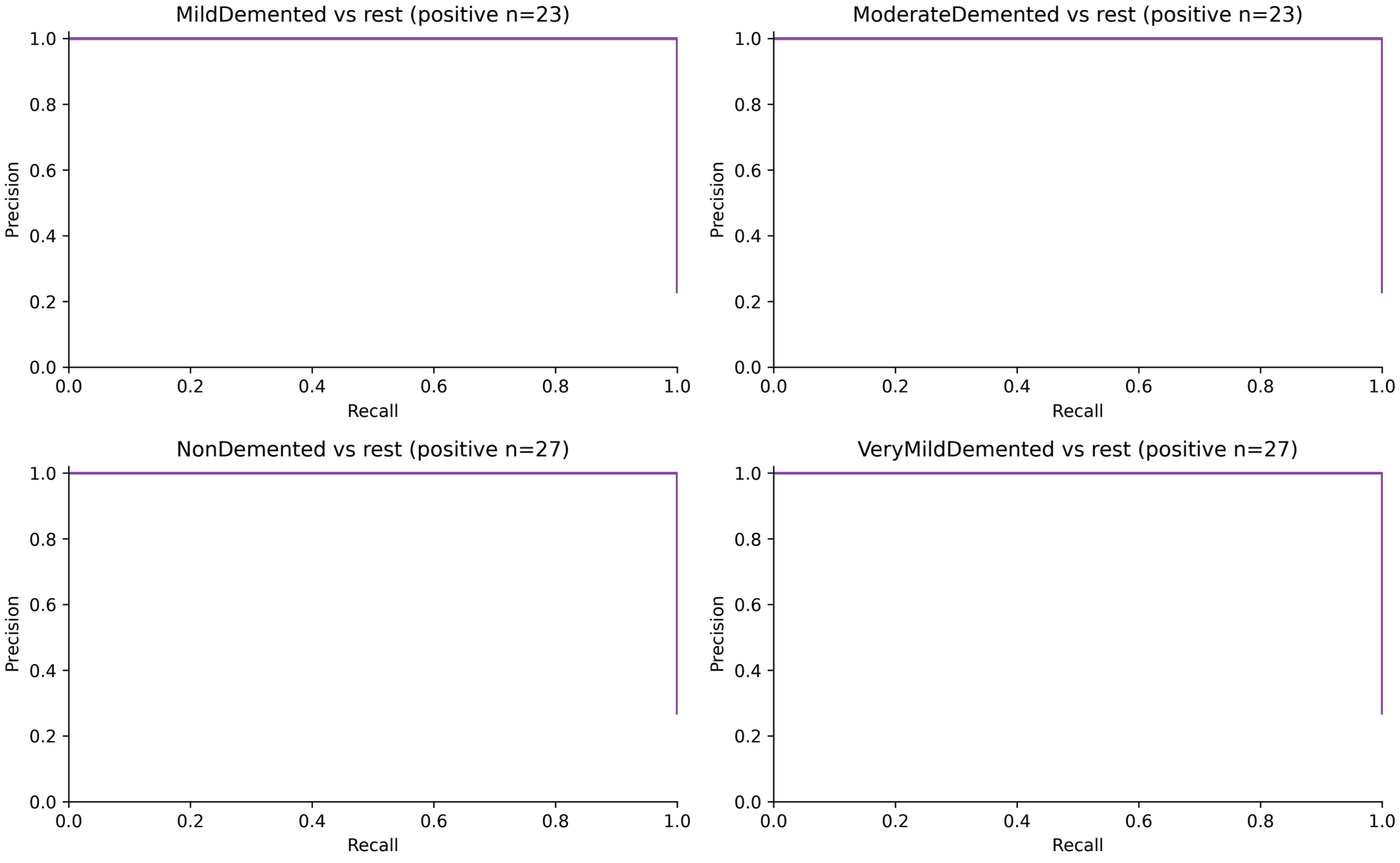


Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

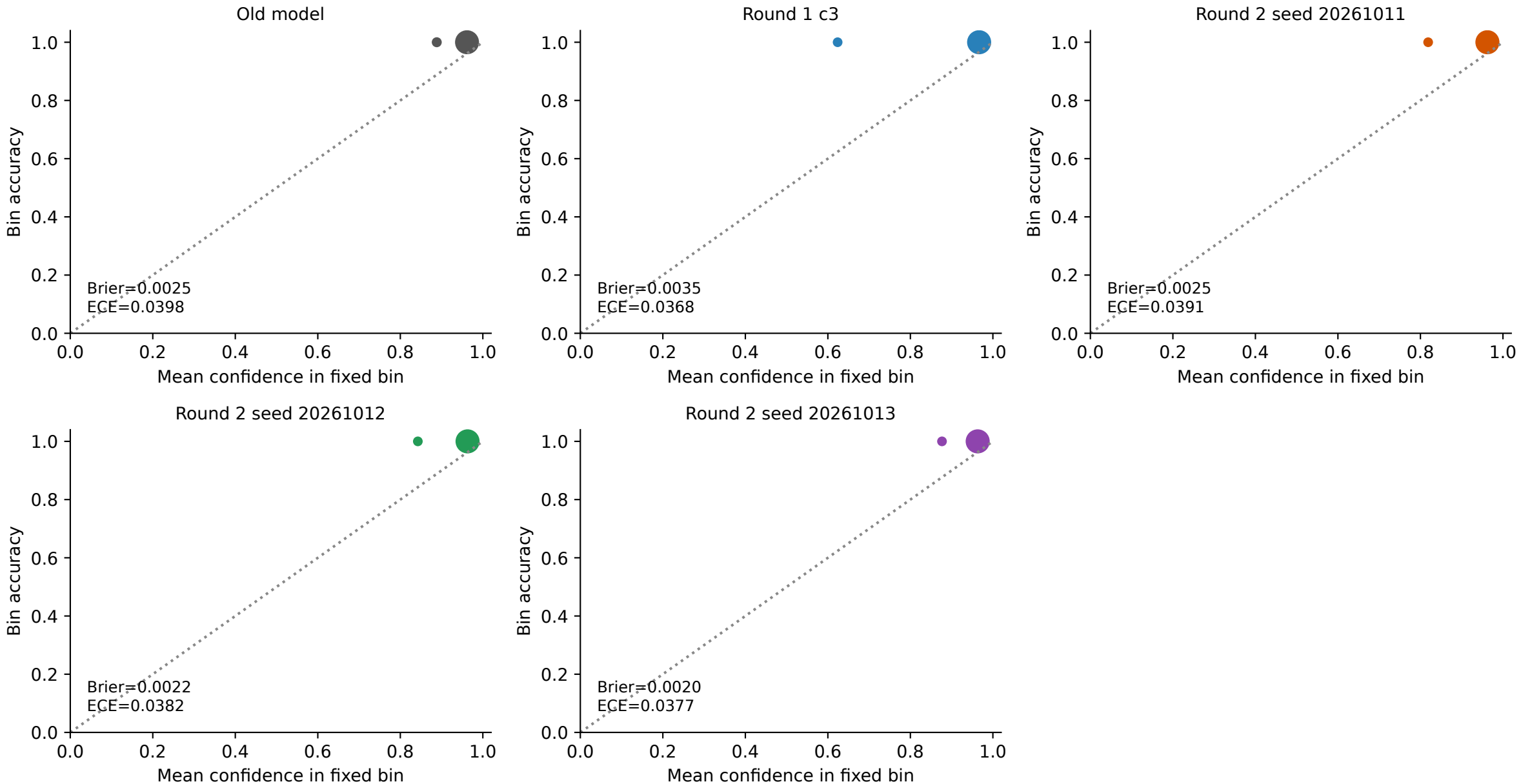

Reused reader benchmark (100) | confusion | n=100
Cells show counts; color shows recall within each reference-class row. All models/readers retain their fixed order.
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Reader Kalapos
Reader Thamburaj
Reader Kanekar
Reference class
Predicted class
Mild
Moderate
Non
Very mild
Reference-row fraction
1.0
0.8
0.6
0.4
0.2
0.0
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.


Reused reader benchmark (100) | fixed stress label flip fraction
brightness
contrast
noise
blur
downsample
jpeg
label flip fraction (%)
Clean
0.9
1.1
0.01
0.02
0.5
192
160
90
70
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Same-device clean reference; all13 unchanged scenarios. Synthetic pixel stress is not MRI acquisition or clinical validation.

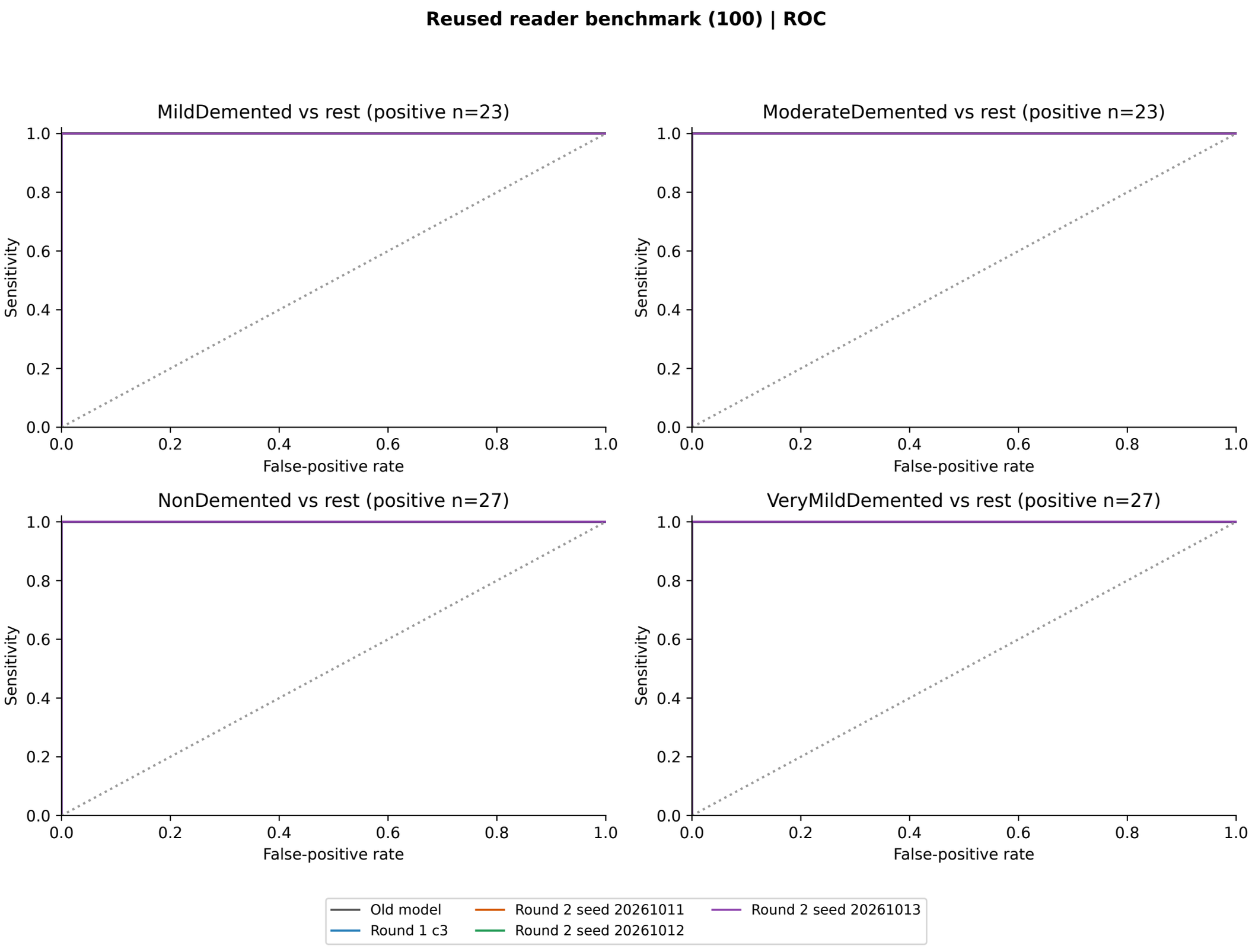
Reused reader benchmark (100) | Precision–recall



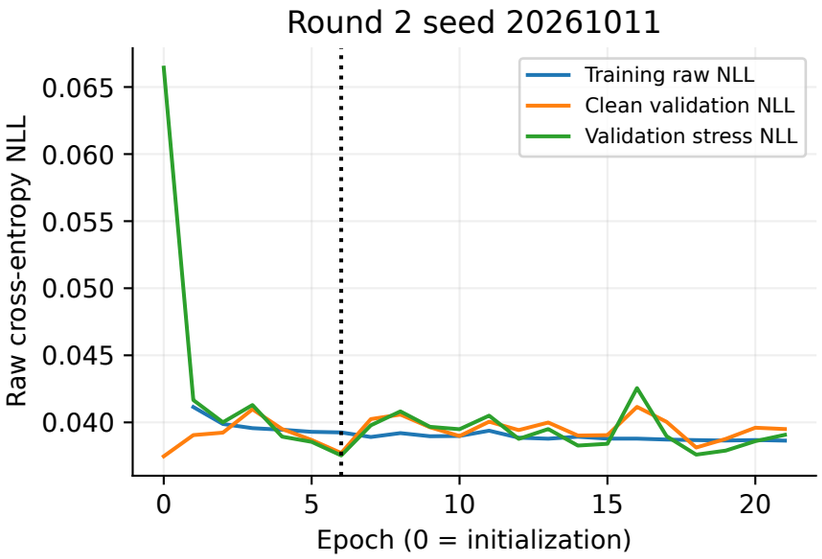
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

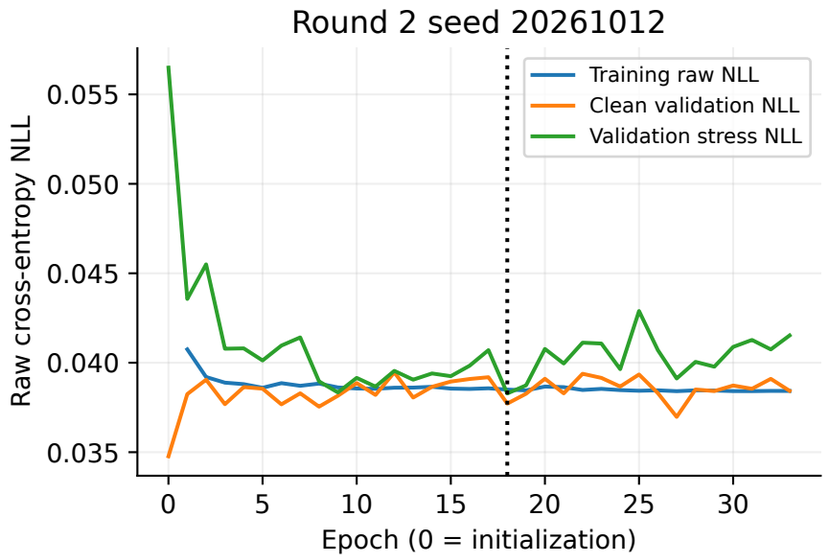


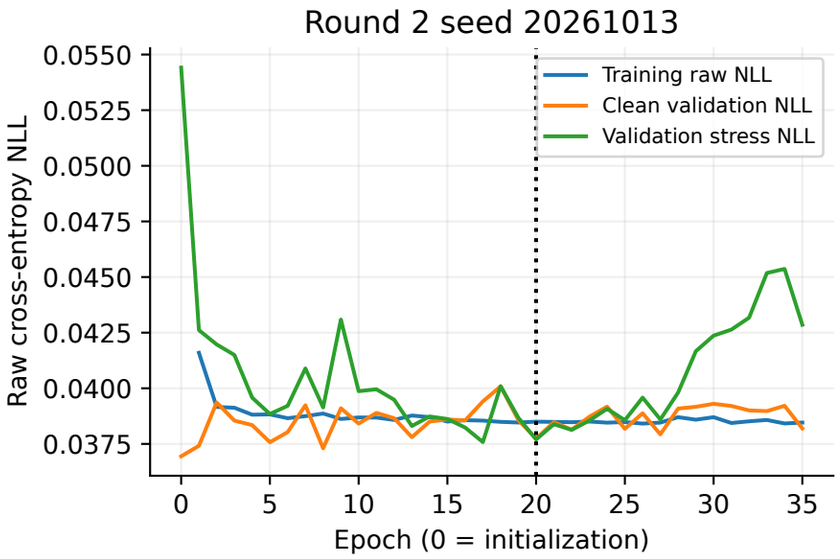
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

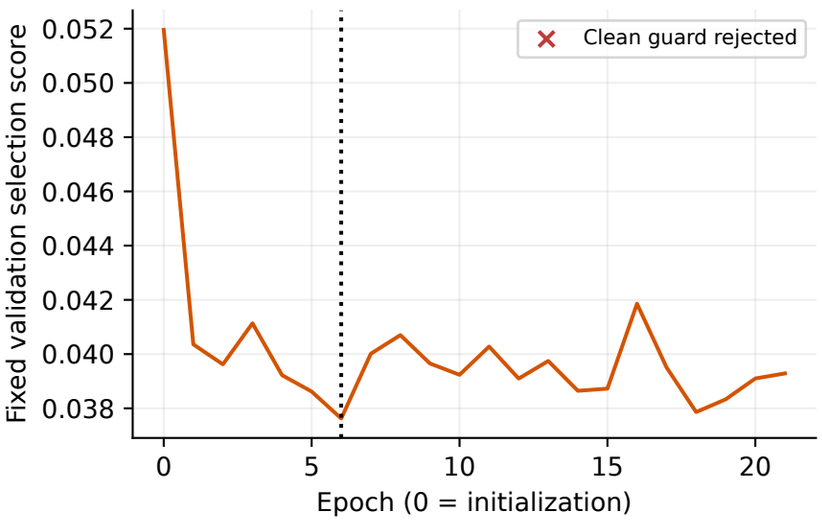

Reused reader benchmark (100) | ROC
MildDemented vs rest (positive n=23)
ModerateDemented vs rest (positive n=23)
NonDemented vs rest (positive n=27)
VeryMildDemented vs rest (positive n=27)
Sensitivity
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Old model
Round 1 c3
Round 2 seed 20261011
Round 2 seed 20261012
Round 2 seed 20261013
Post-hoc repeated image benchmarks; hash groups are not patients. No clinical or independent-test claim.

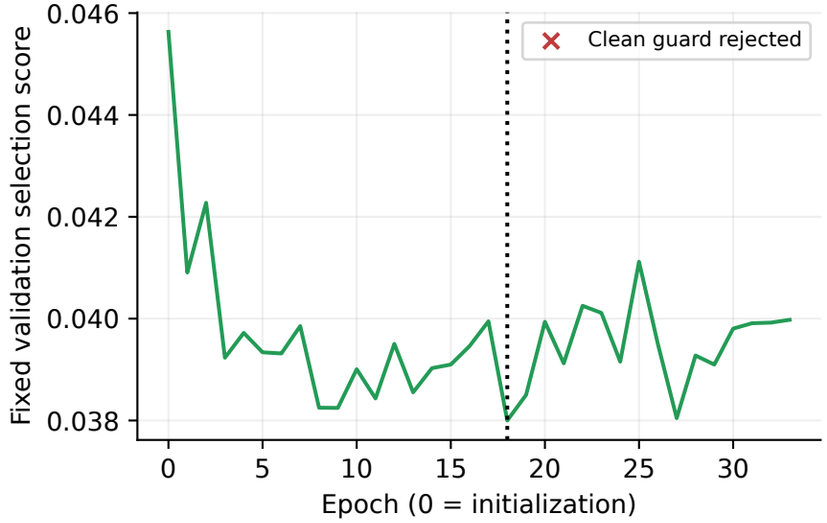
Round2 raw learning curves | all three runs, selected epoch marked

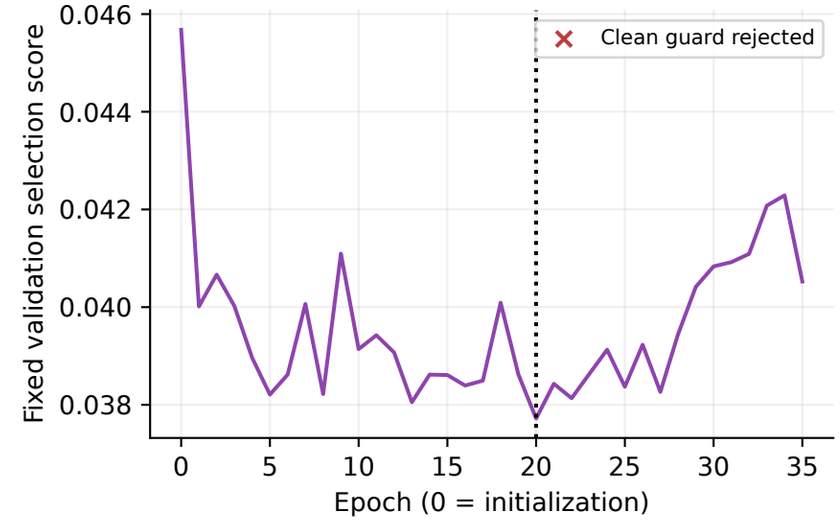


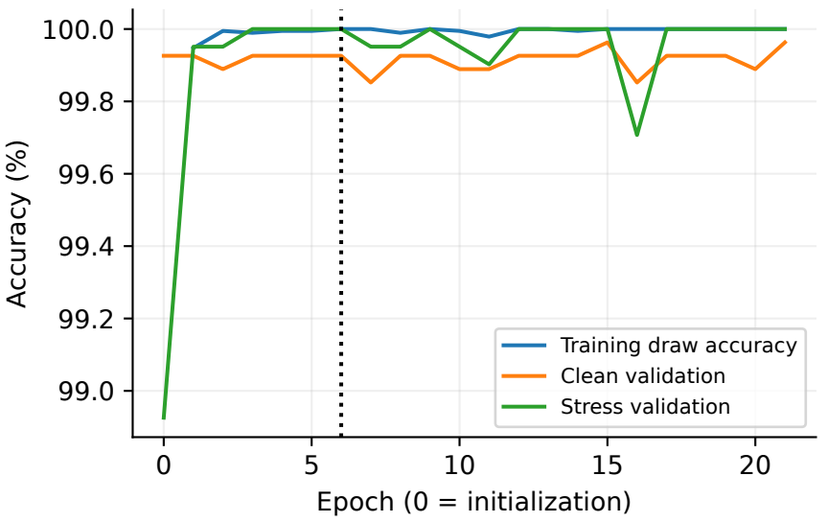


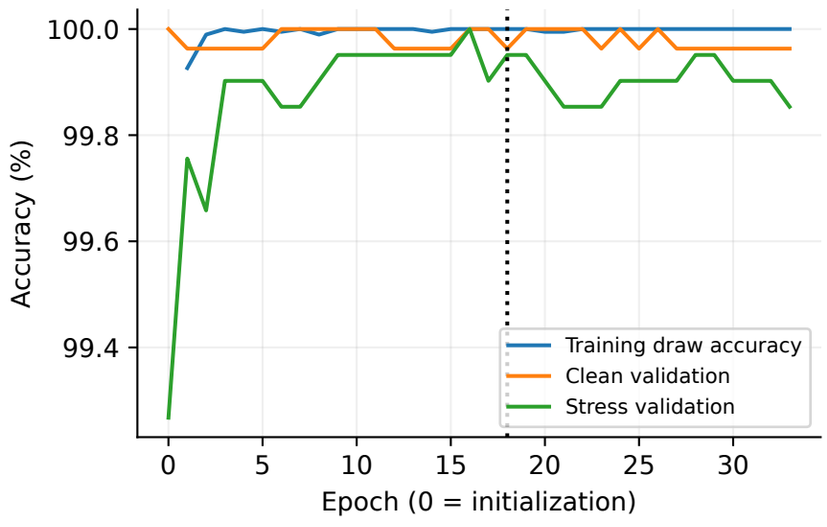


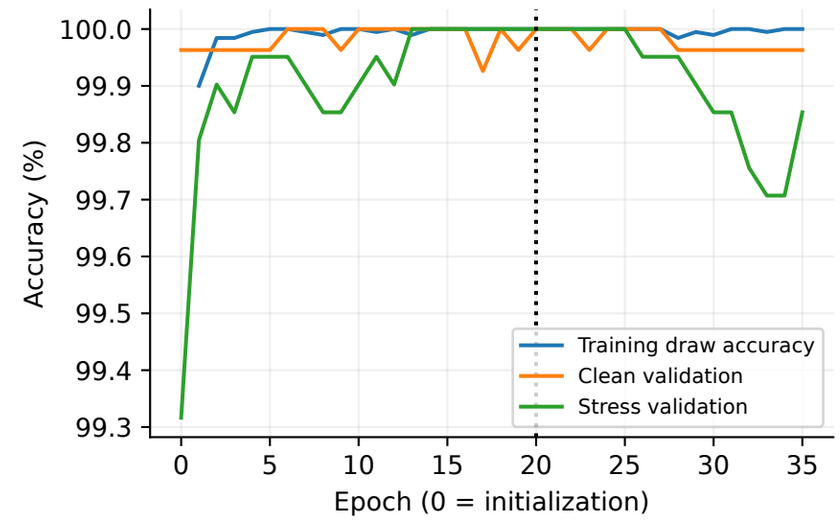




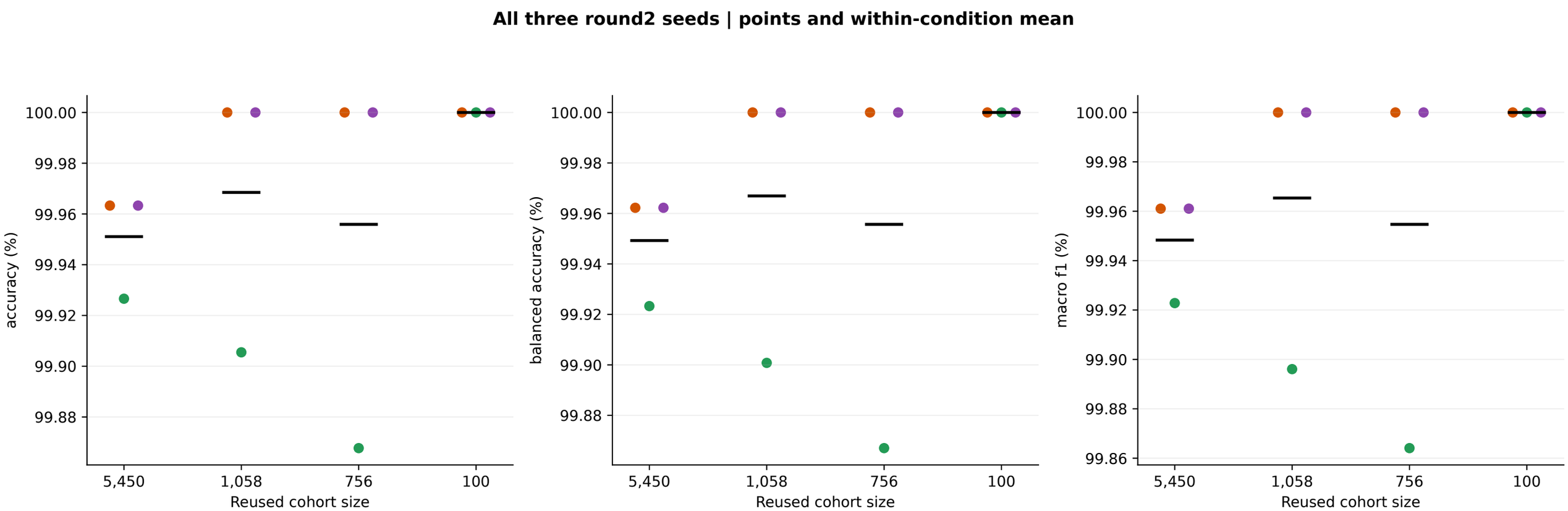




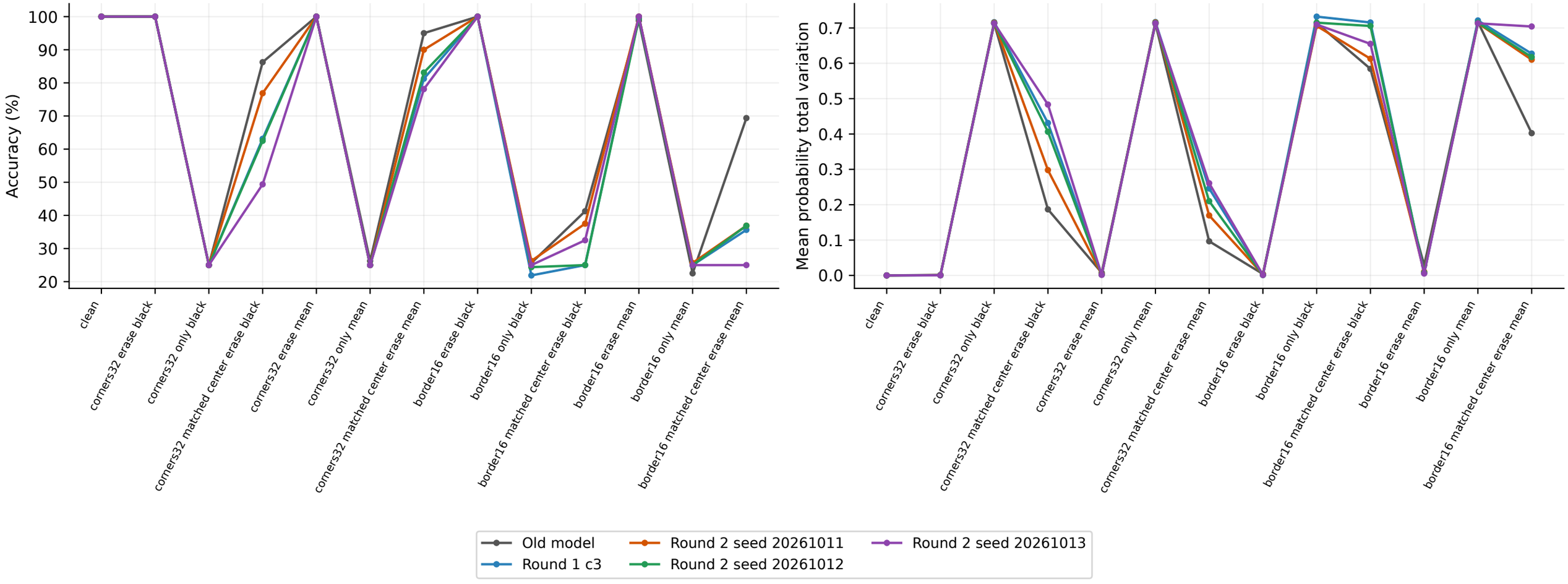




Training includes random augmentation; validation is fixed. Raw NLL differs from the weighted, label-smoothed optimization objective.
Epoch0 fallback, if selected, is retained; longer training is not evidence of a better model.



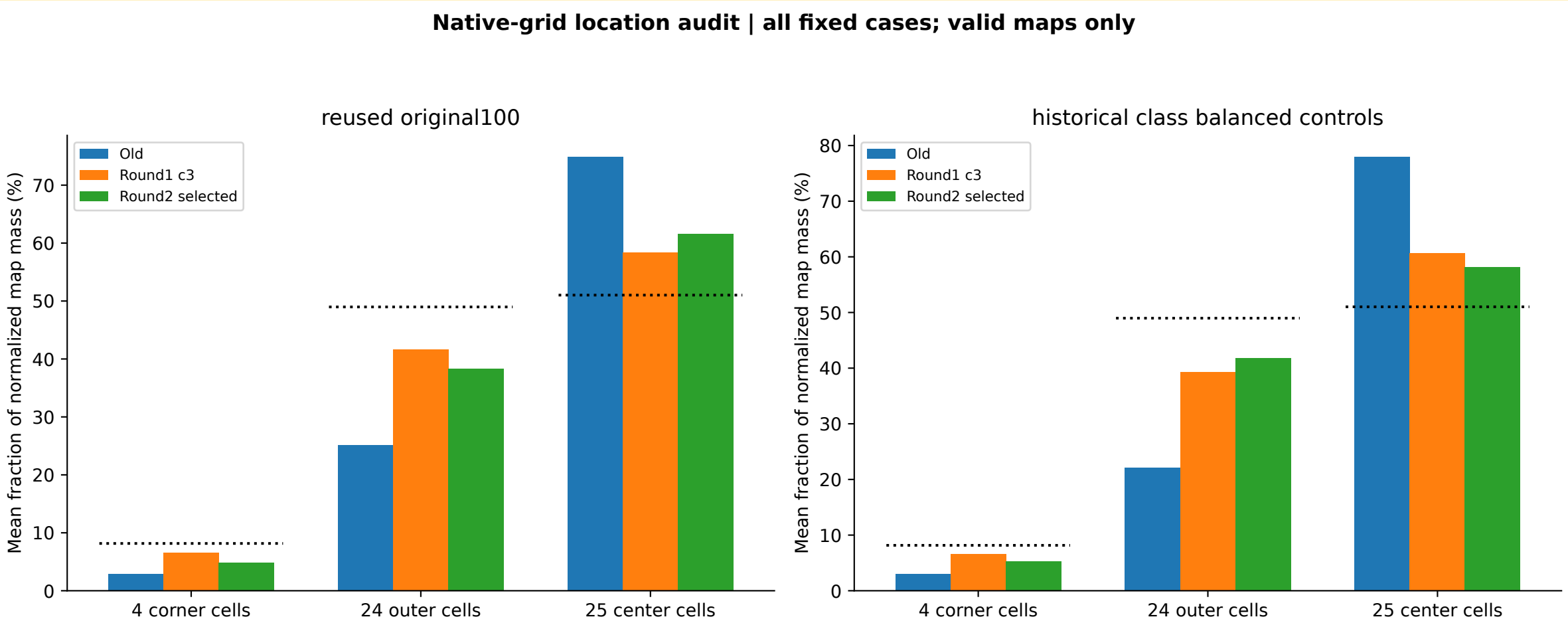
Seed variation is descriptive; the validation representative is fixed independently of these outcomes.



Erasure is out of distribution. Matched center areas and both fills are controls; these results cannot exclude all shortcuts.

DIAGNOSTIC ONLY — probability replay check failed (2/100; classes unchanged).

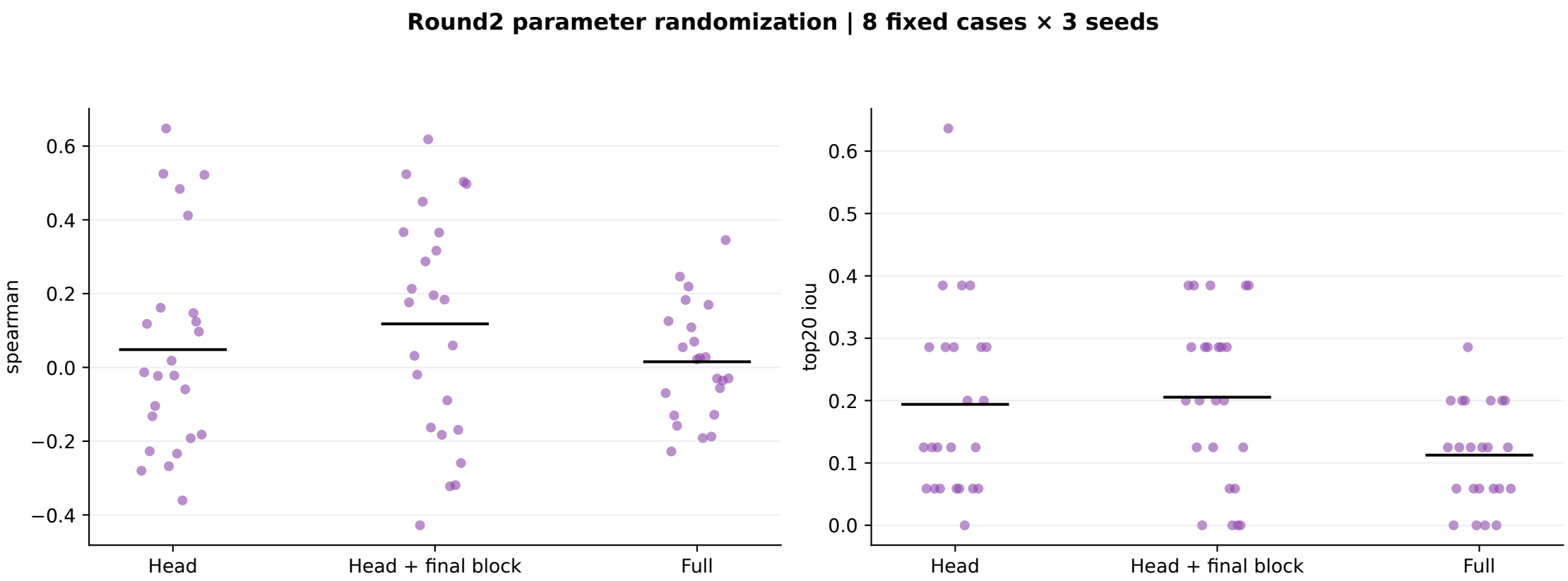
Tolerance 0.0001 unchanged. Not anatomical or clinical validation. See replay limitation.



Dotted lines: uniform area fractions. Outer cells overlap corner cells. Map mass is not anatomical probability or causal input importance.

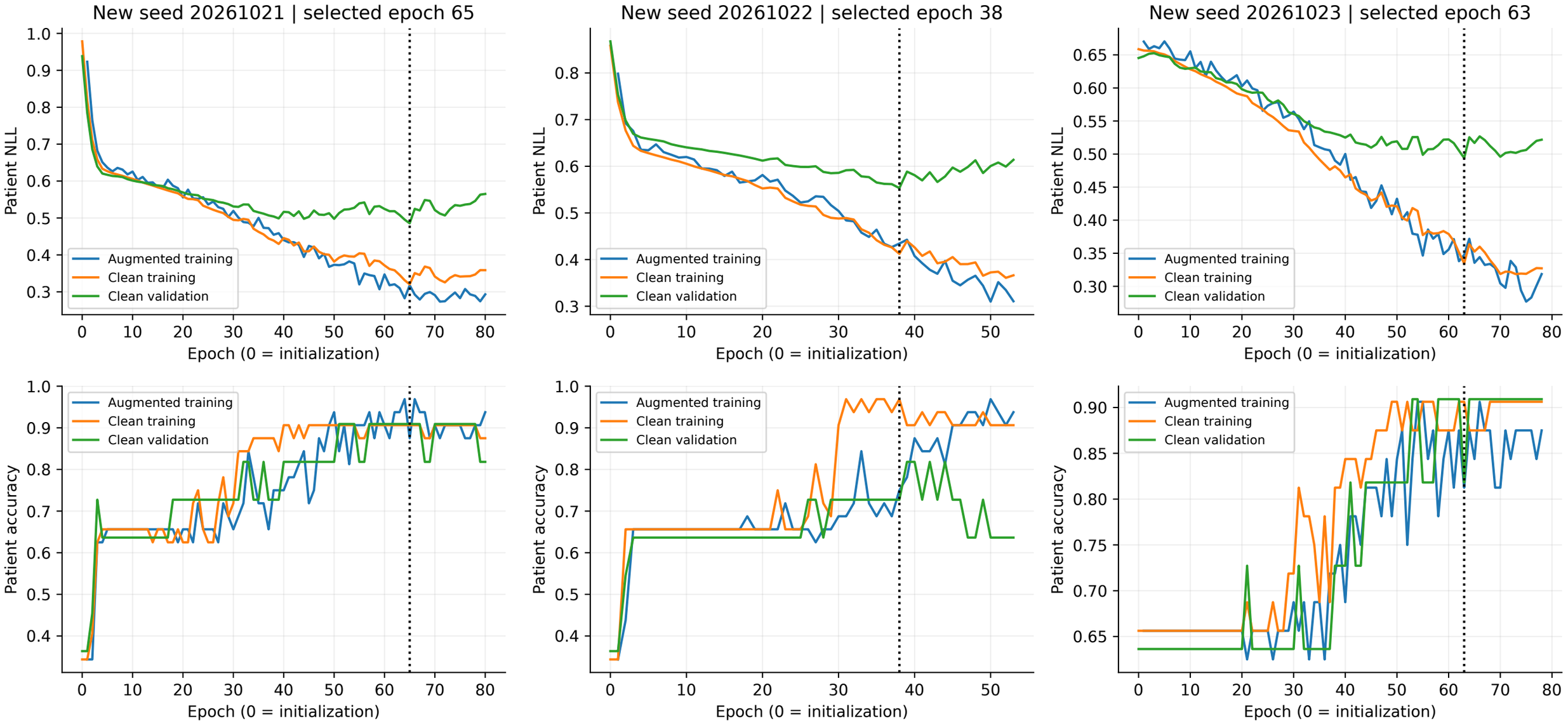
DIAGNOSTIC ONLY — probability replay check failed (2/100; classes unchanged).

Tolerance 0.0001 unchanged. Not anatomical or clinical validation. See replay limitation.



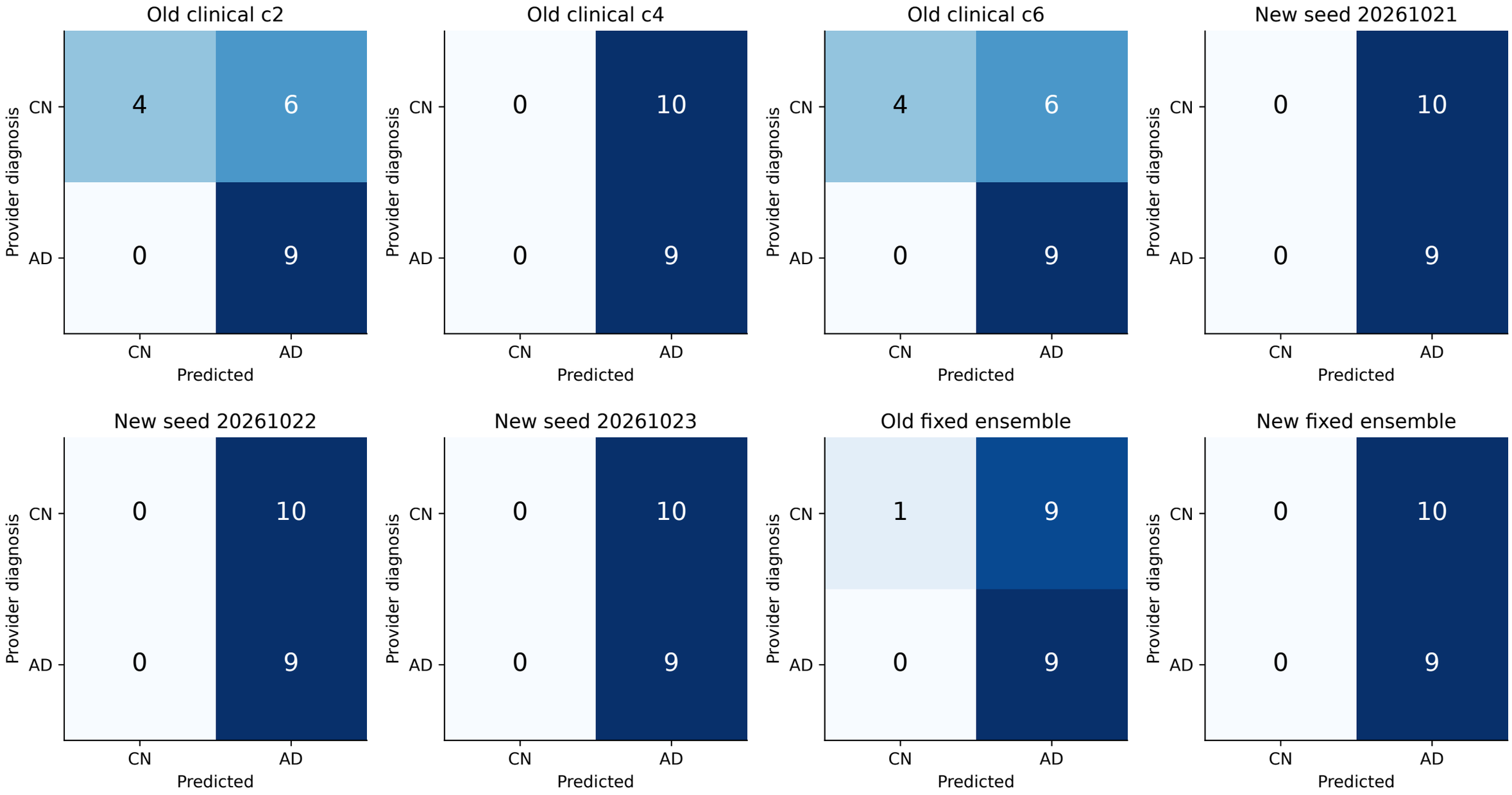
Original predicted class stays fixed. Points exclude flat/failed maps, whose counts remain in randomization.csv. Model state is restored.

All three clinical runs | raw unsmoothed training and validation curves

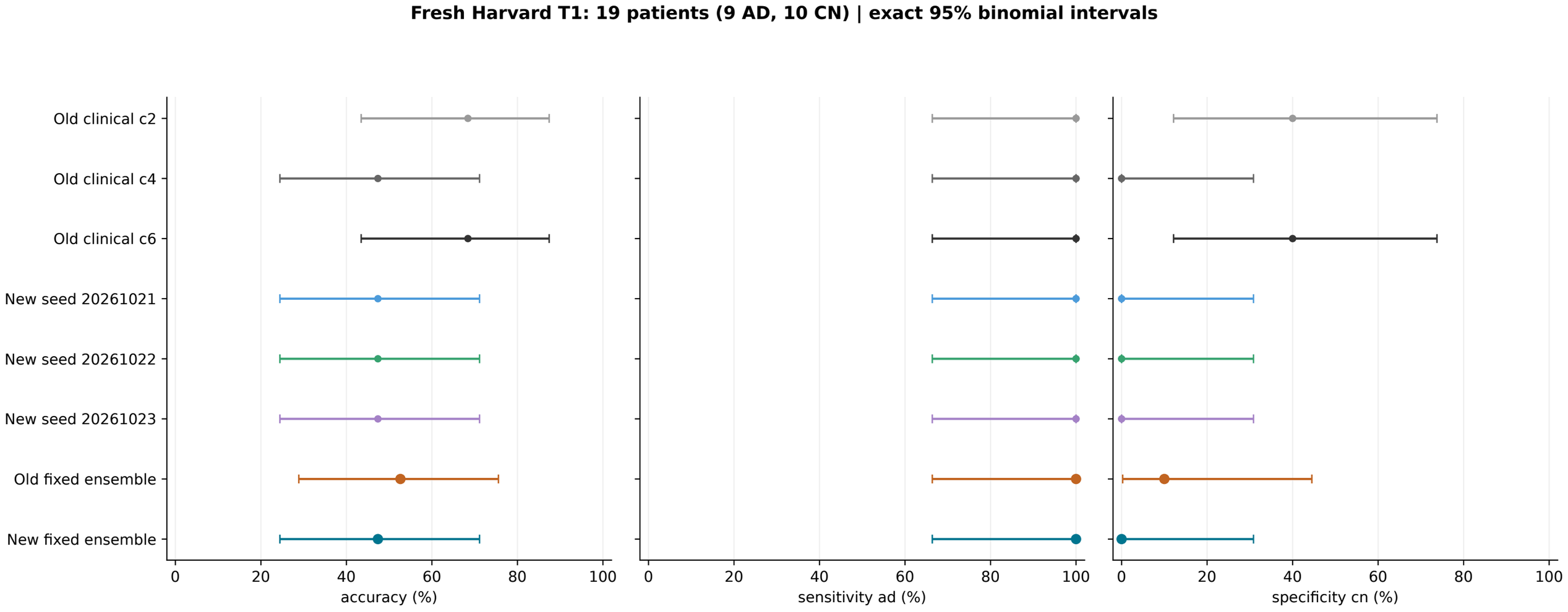


32 training and 11 validation patients. Clean training is resubstitution; augmentation is applied only to optimization draws.
Stored trainer NLL uses its fixed 1e-15 clipping; held-out NLL uses the frozen comparison helper 1e-12 clipping.

Fresh Harvard T1: 19 patients (9 AD, 10 CN) | patient confusion counts

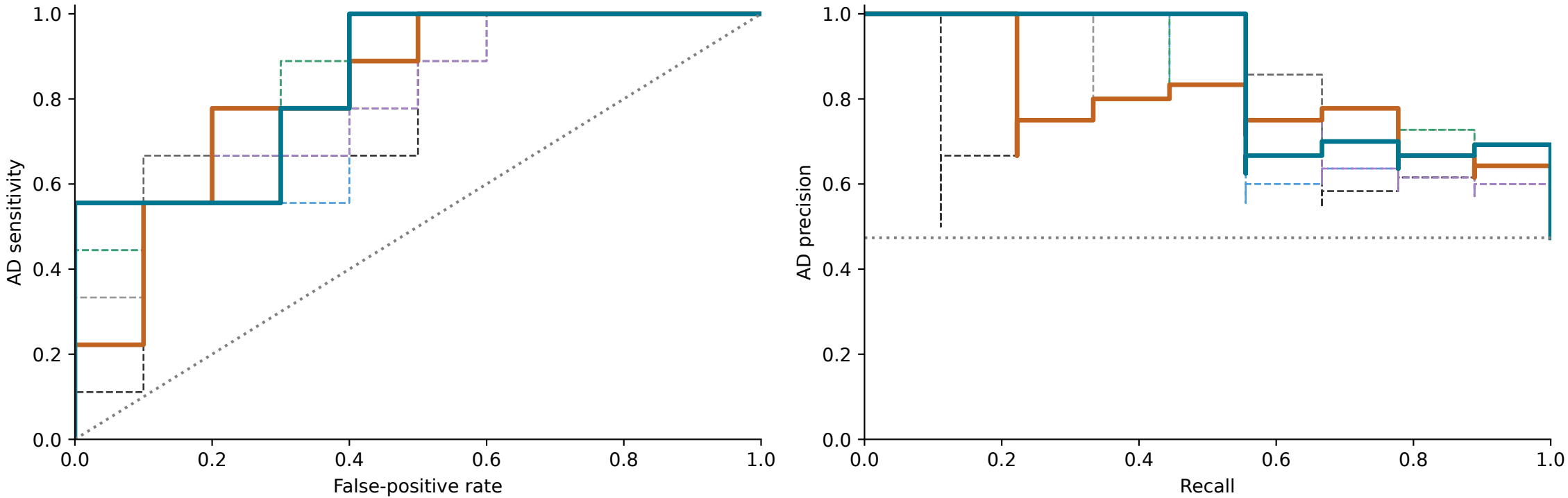


Patient-level research pilot; small convenience cohorts. No clinical superiority or population risk-calibration claim.



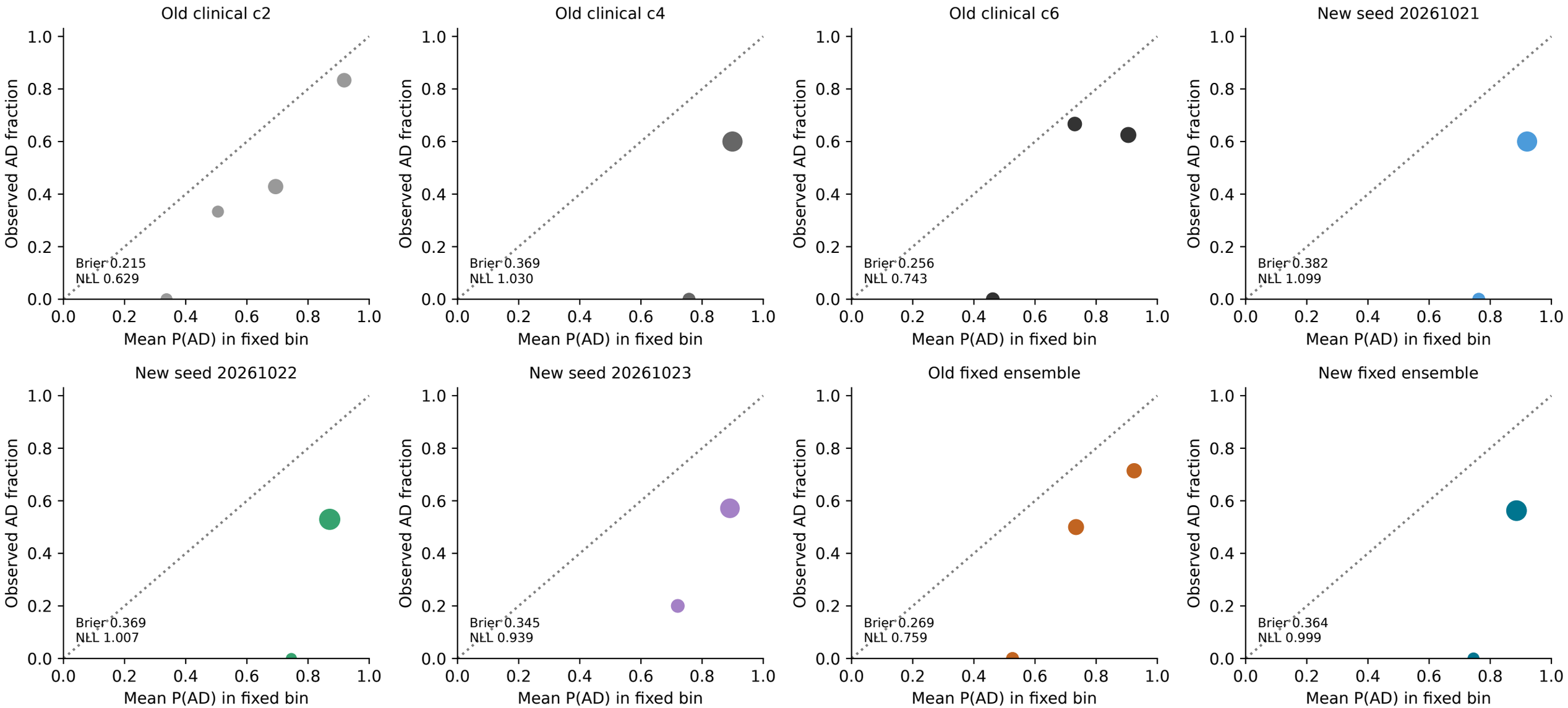
Patient-level research pilot; small convenience cohorts. No clinical superiority or population risk-calibration claim.
Separate sensitivity/specificity denominators; seed and patient uncertainty are distinct.

Fresh Harvard T1: 19 patients (9 AD, 10 CN) | ROC and precision–recall



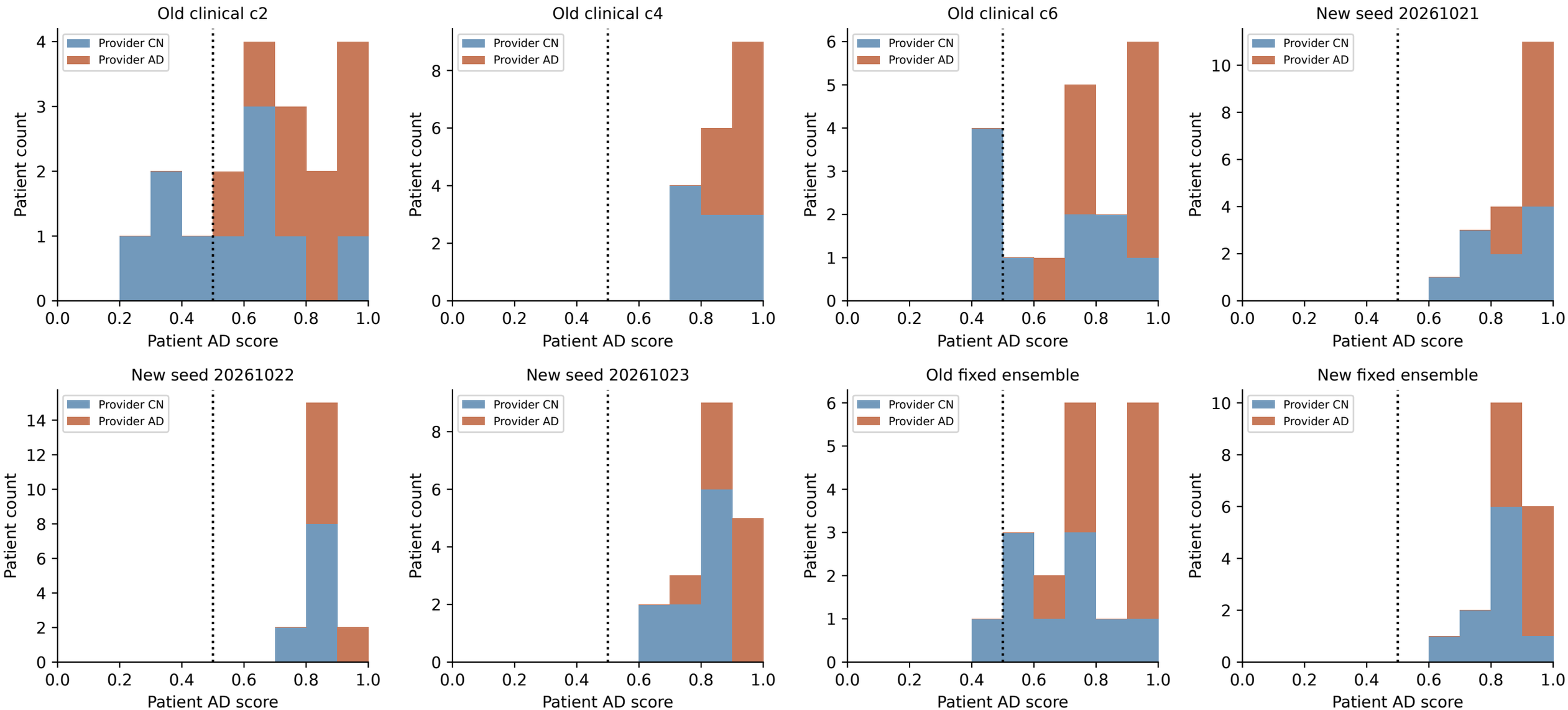
Small fixed case-control samples; precision reflects sampled prevalence. All scores and uncertainty appear in analysis.json.

Fresh Harvard T1: 19 patients (9 AD, 10 CN) | reliability



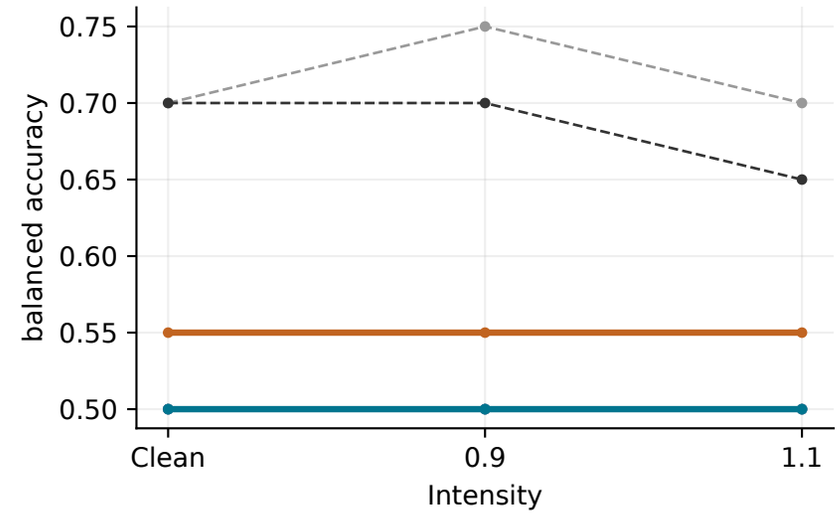
Five fixed probability bins are descriptive; sparse bins and case-control prevalence do not establish clinical-risk calibration.

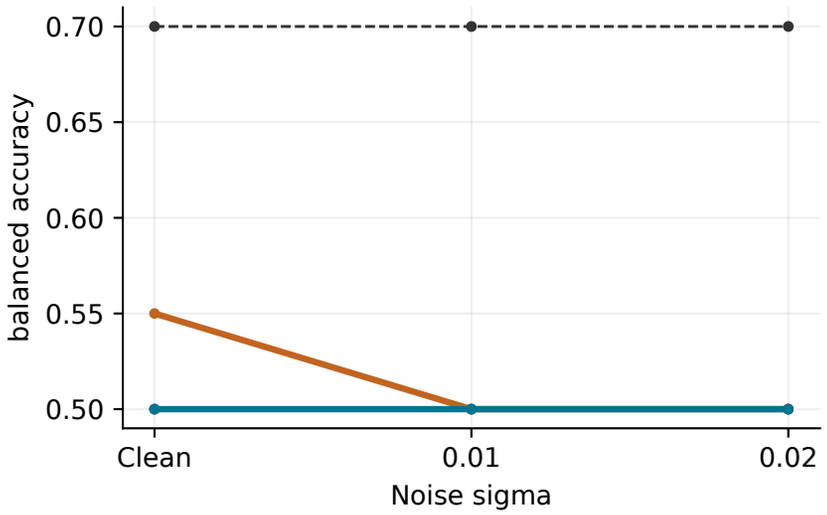
Fresh Harvard T1: 19 patients (9 AD, 10 CN) | confidence

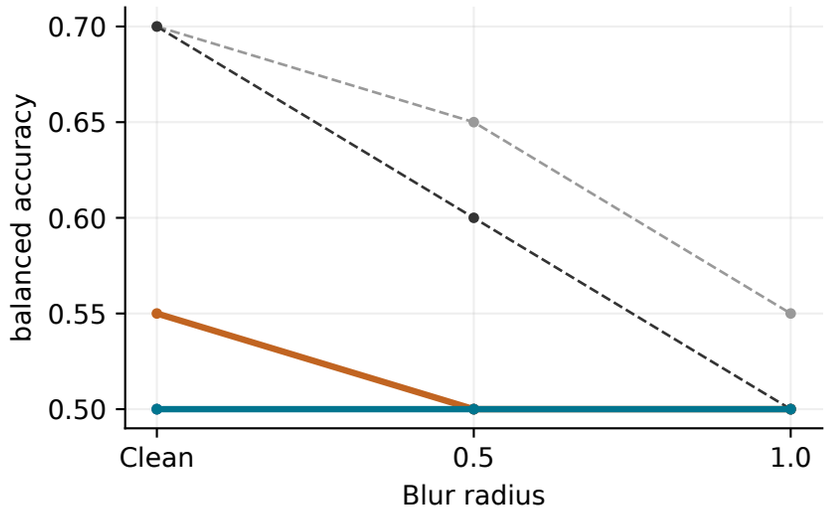


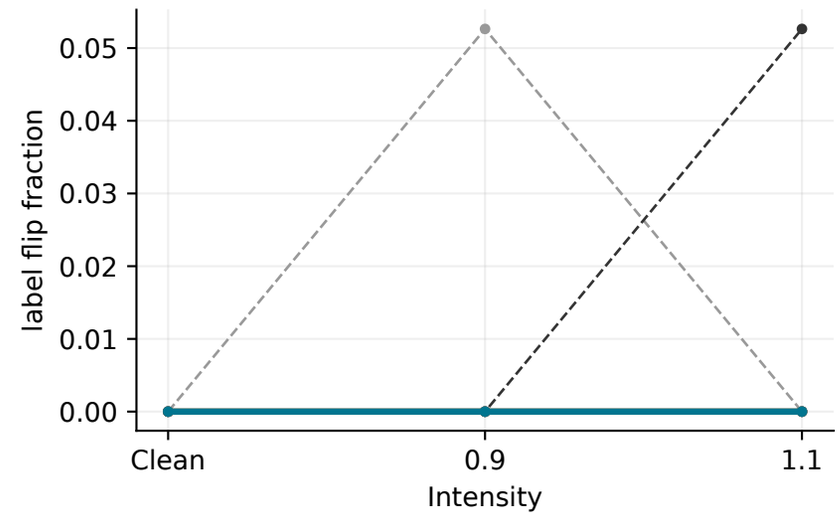
Ten fixed histogram bins show patient scores by provider diagnosis; case-control score distributions do not establish clinical-risk calibration.

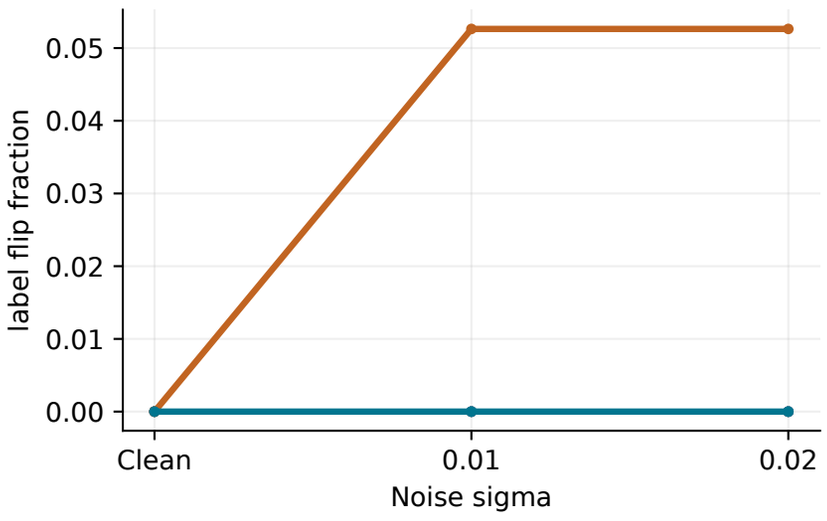
Fresh Harvard T1: 19 patients (9 AD, 10 CN) | all 7 fixed pixel-stress conditions

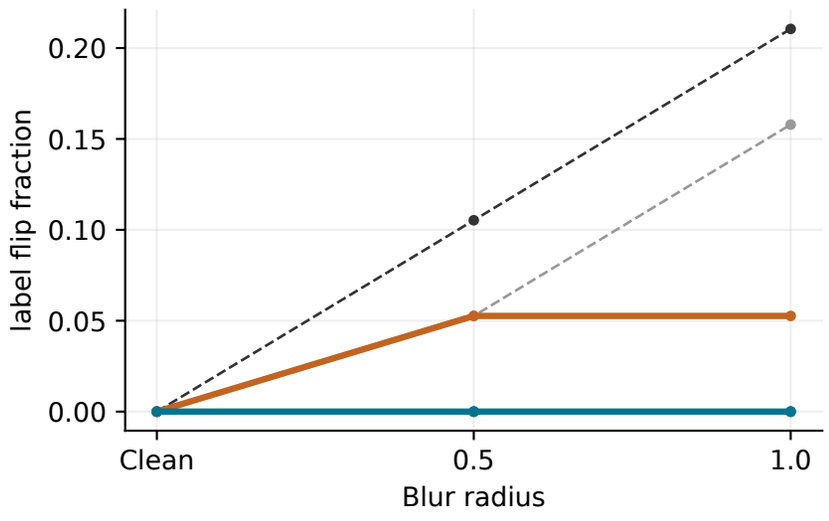


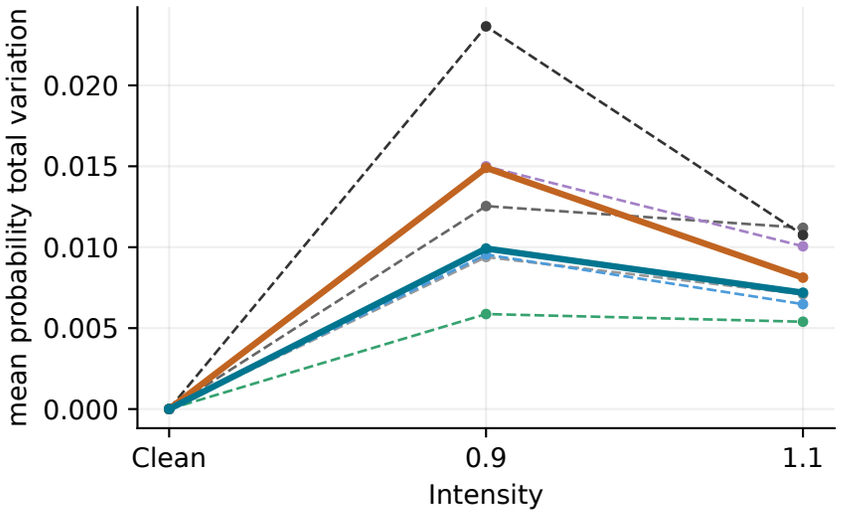


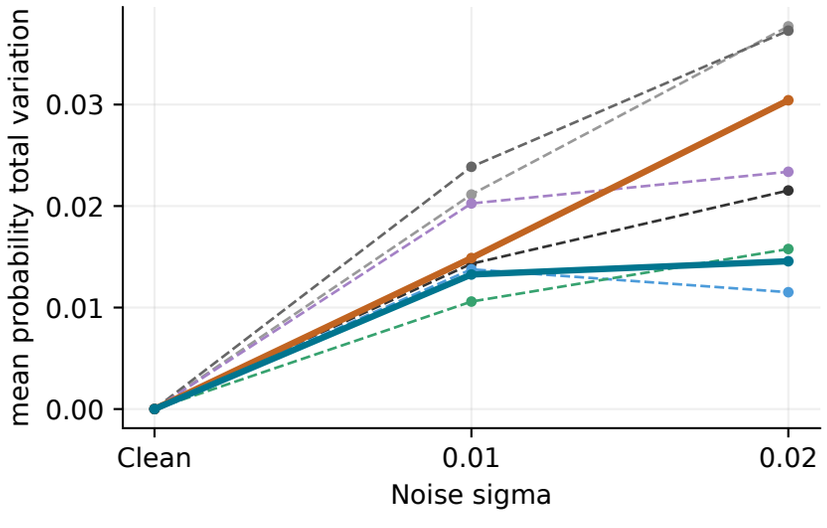


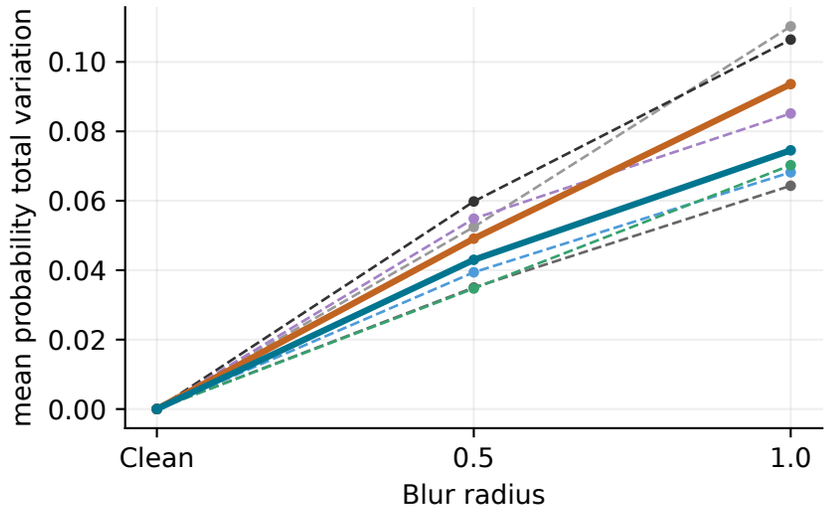




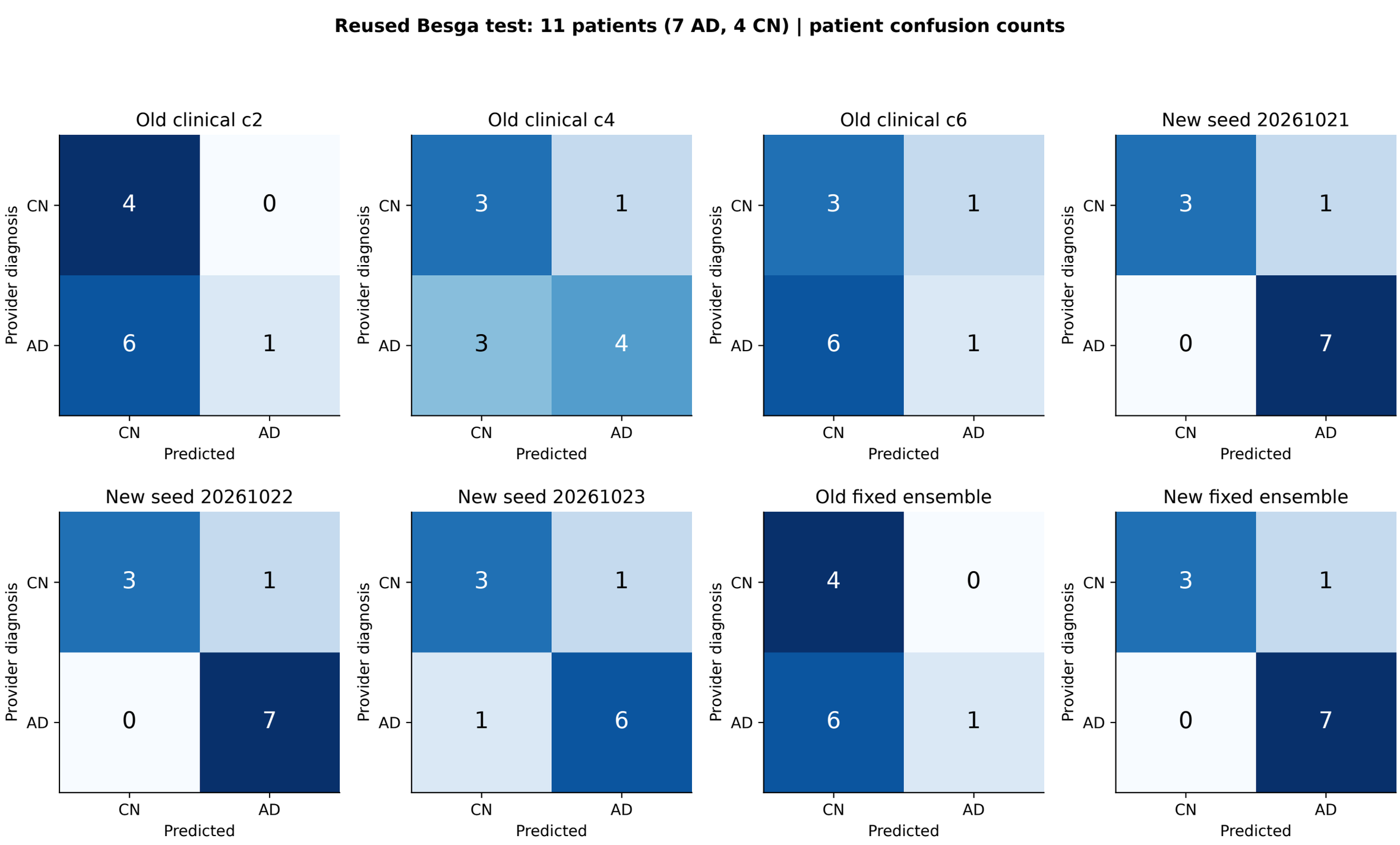


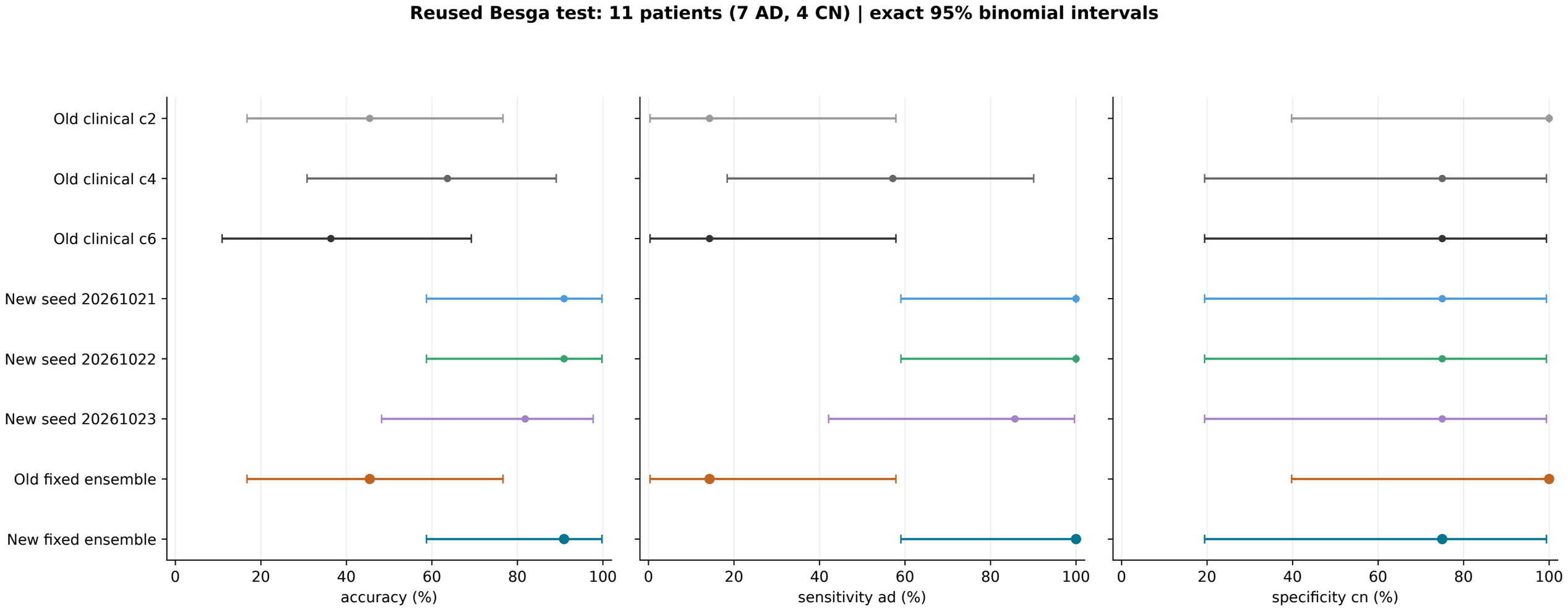




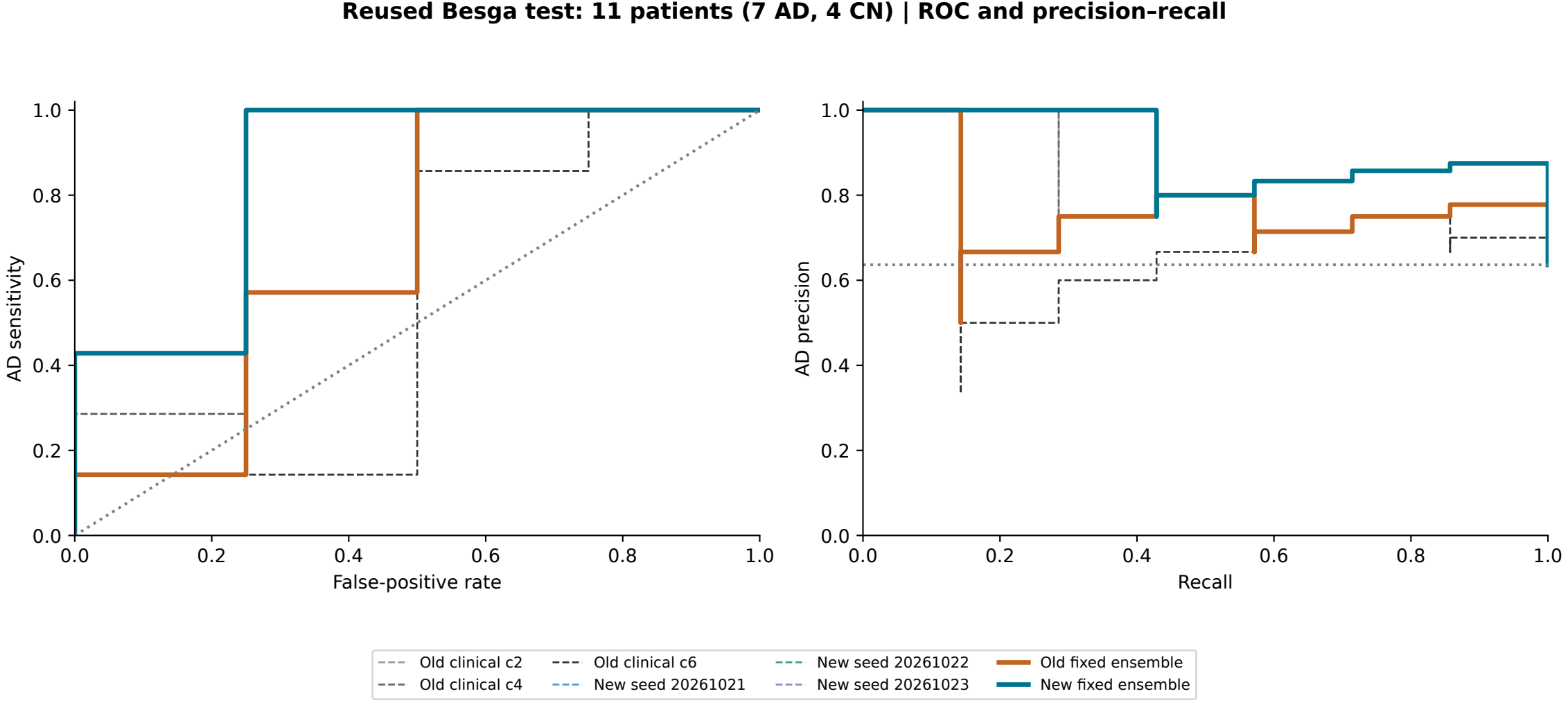


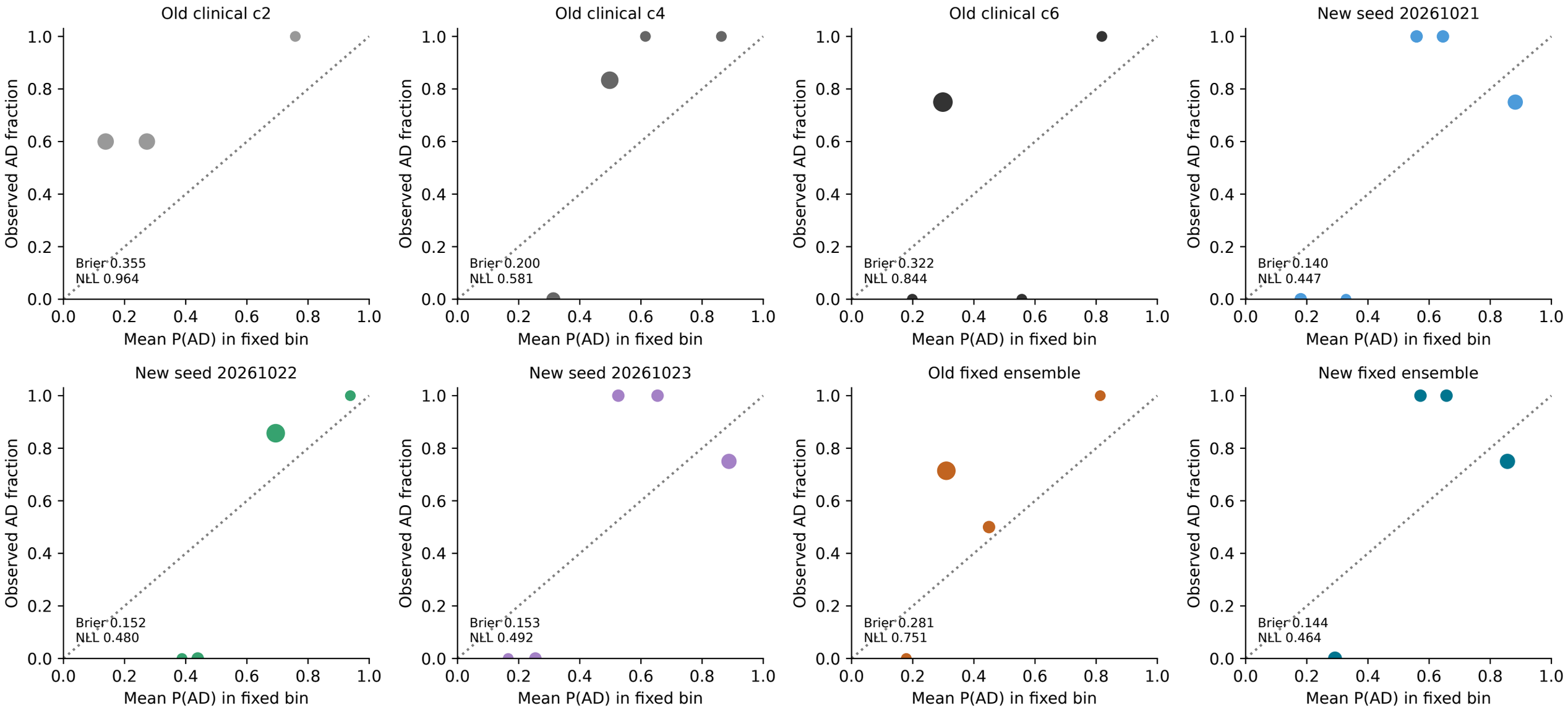
Same patient, perturbation and clean reference for every model. These synthetic 2D controls do not simulate MRI acquisition physics.


Reused Besga test: 11 patients (7 AD, 4 CN) | patient confusion counts
Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble
Provider diagnosis
CN
AD
Predicted
4
0
6
1
3
1
3
4
3
1
6
1
3
1
0
7
3
1
0
7
3
1
1
6
4
0
6
1
3
1
0
7
Patient-level research pilot; small convenience cohorts. No clinical superiority or population risk-calibration claim.



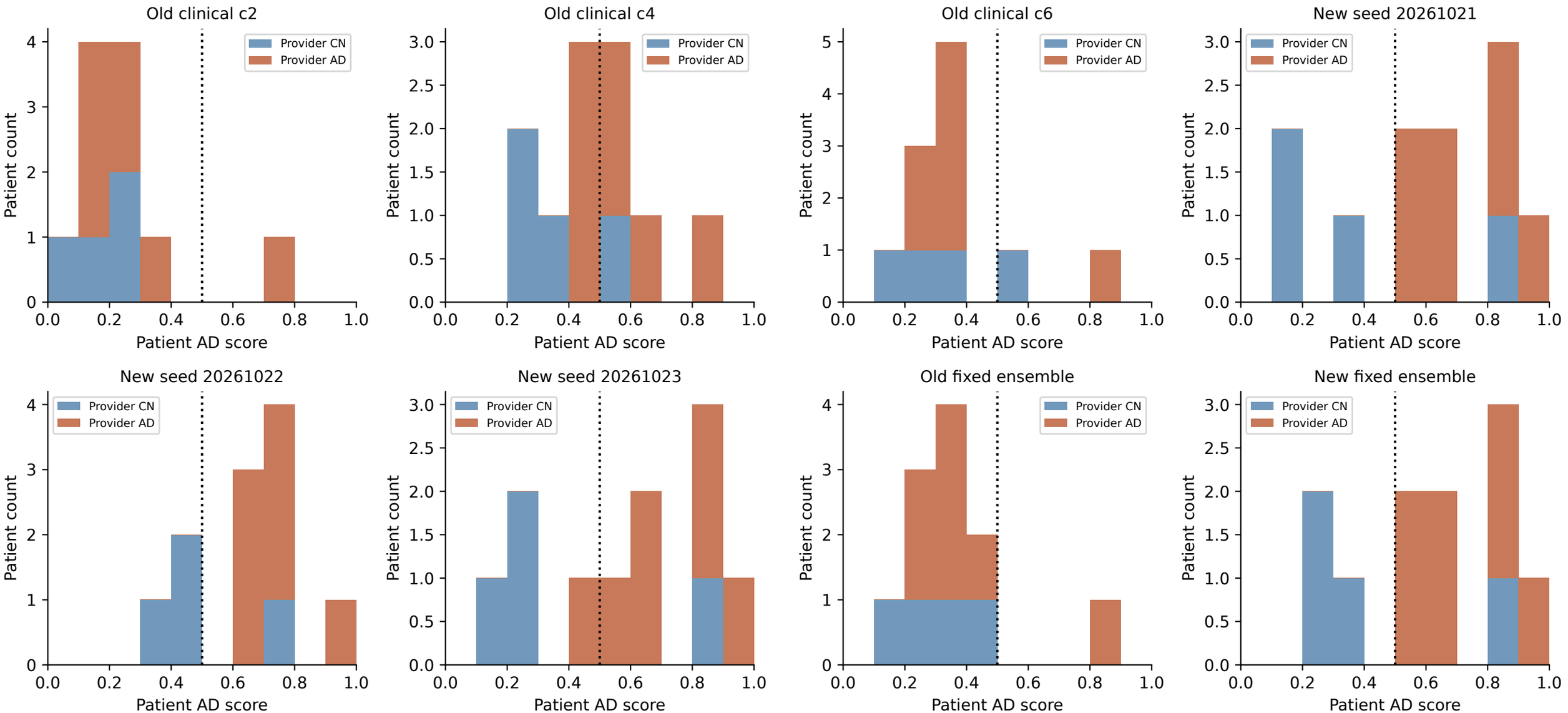
Patient-level research pilot; small convenience cohorts. No clinical superiority or population risk-calibration claim.
Separate sensitivity/specificity denominators; seed and patient uncertainty are distinct.


Reused Besga test: 11 patients (7 AD, 4 CN) | ROC and precision–recall
AD sensitivity
False-positive rate
AD precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0
Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble
Small fixed case-control samples; precision reflects sampled prevalence. All scores and uncertainty appear in analysis.json.



Five fixed probability bins are descriptive; sparse bins and case-control prevalence do not establish clinical-risk calibration.

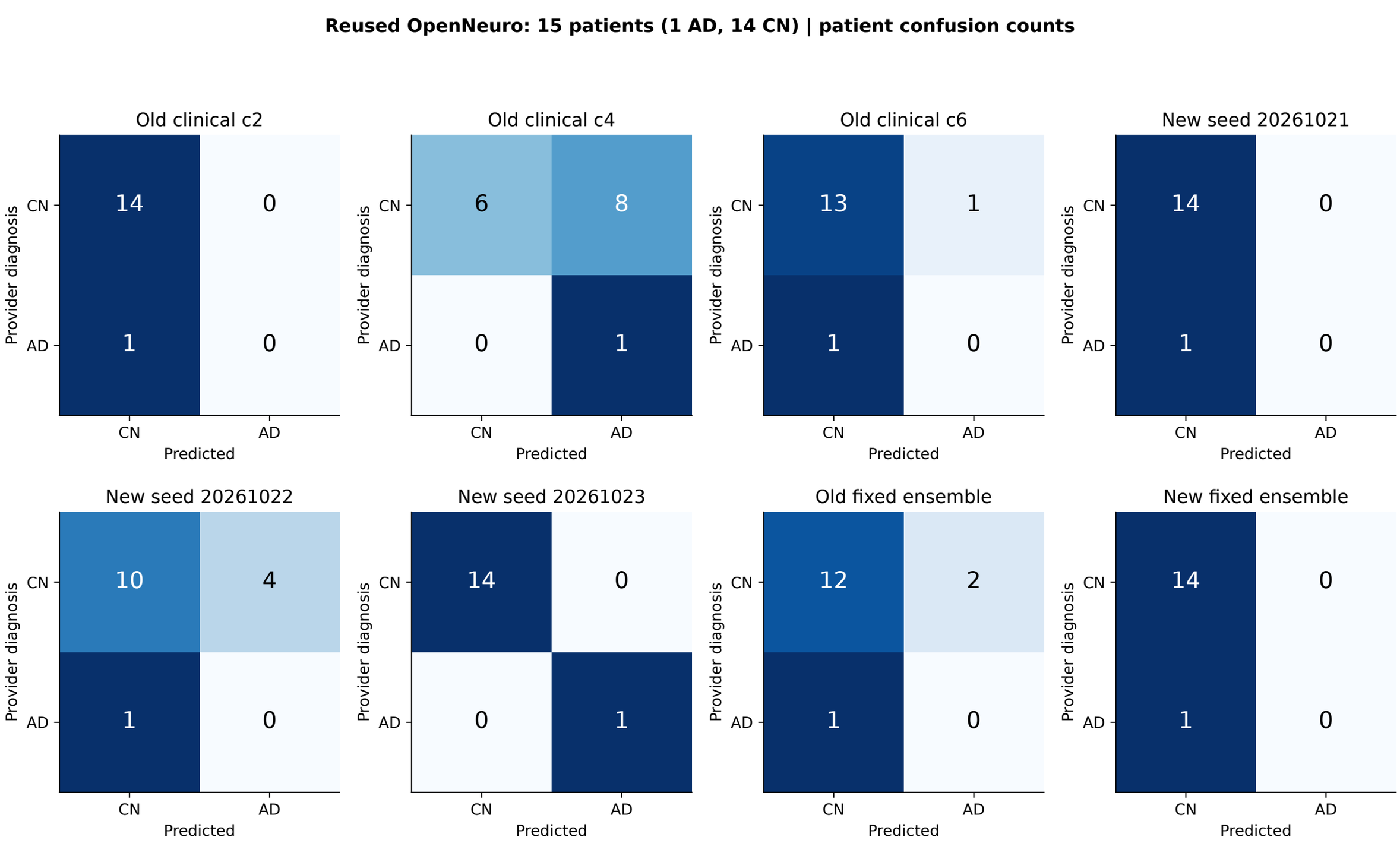
Reused Besga test: 11 patients (7 AD, 4 CN) | confidence

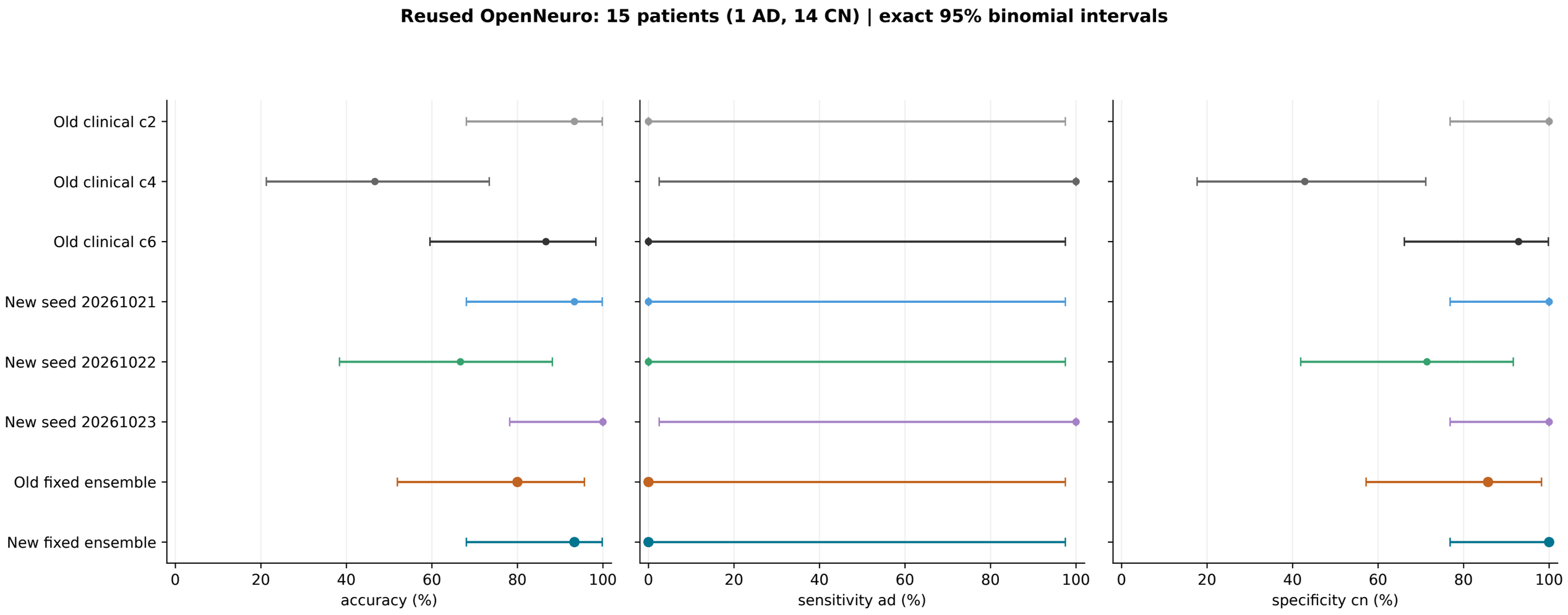


Ten fixed histogram bins show patient scores by provider diagnosis; case-control score distributions do not establish clinical-risk calibration.

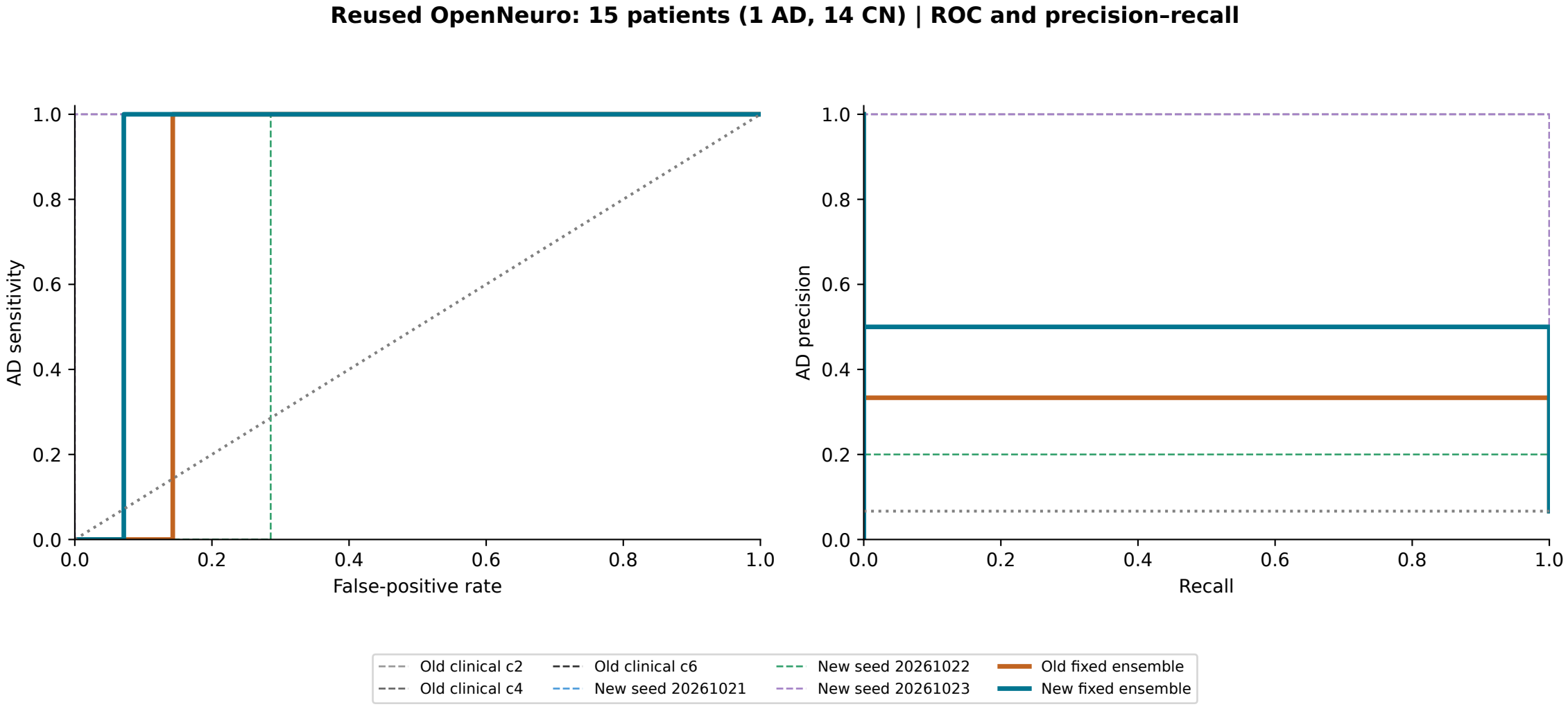
Reused Besga test: 11 patients (7 AD, 4 CN) | all 7 fixed pixel-stress conditions

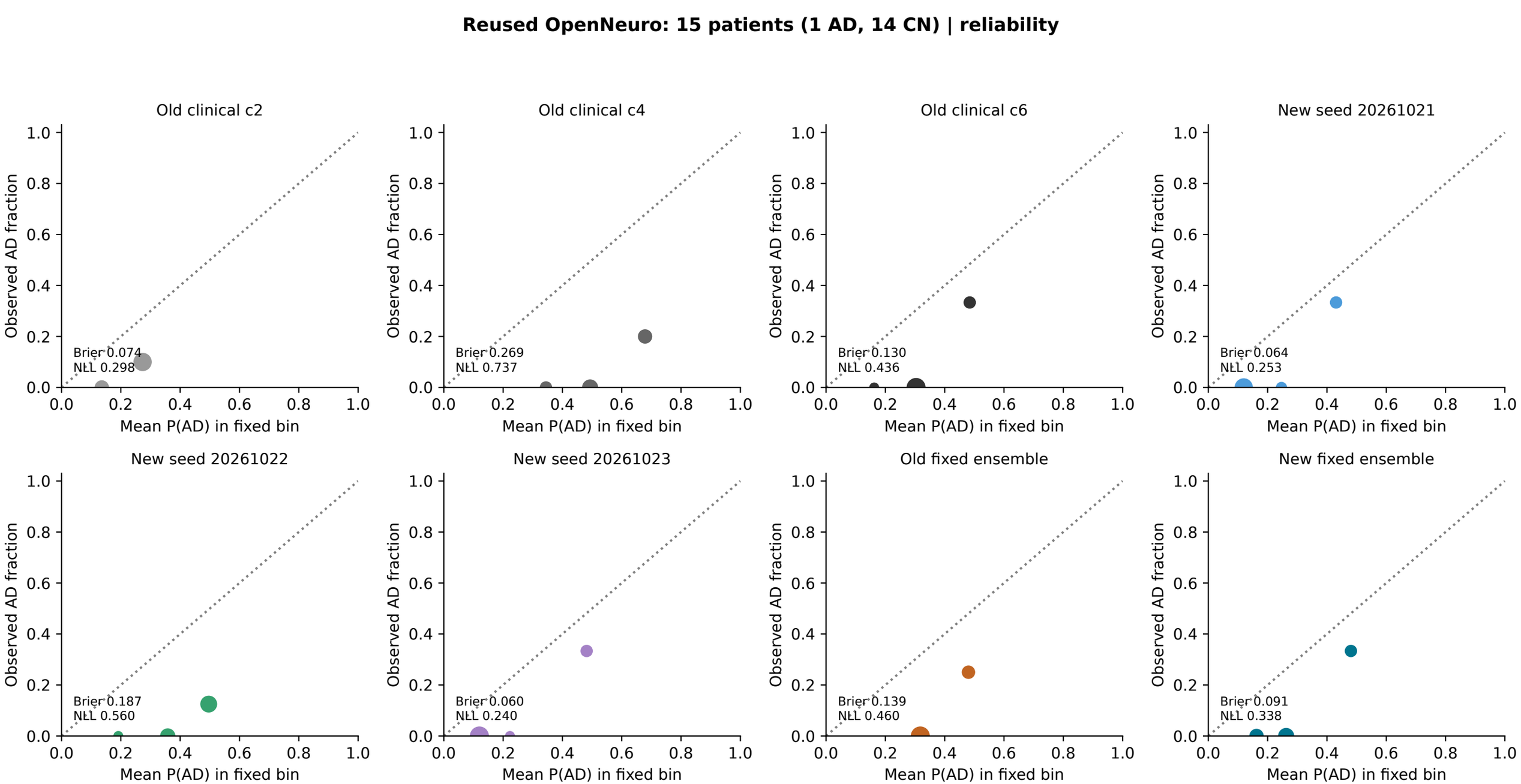
Same patient, perturbation and clean reference for every model. These synthetic 2D controls do not simulate MRI acquisition physics.


Reused OpenNeuro: 15 patients (1 AD, 14 CN) | patient confusion counts
Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble
Provider diagnosis
CN
AD
Predicted
14
0
1
0
6
8
0
1
13
1
1
0
14
0
1
0
10
4
1
0
14
0
0
1
12
2
1
0
14
0
1
0
Only one AD patient: exact sensitivity intervals remain broad; conditional bootstrap does not establish population sensitivity precision.

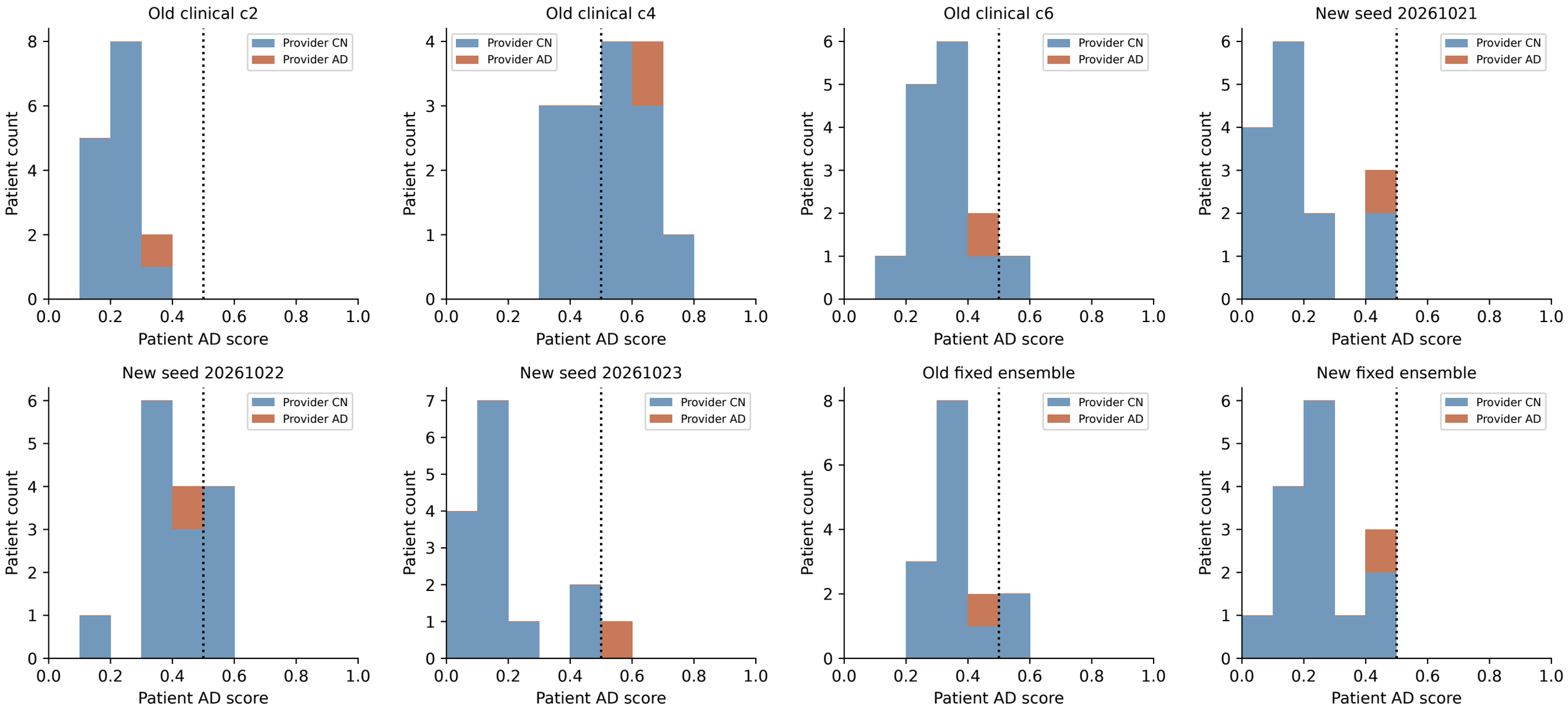


Only one AD patient: exact sensitivity intervals remain broad; conditional bootstrap does not establish population sensitivity precision.
Separate sensitivity/specificity denominators; seed and patient uncertainty are distinct.

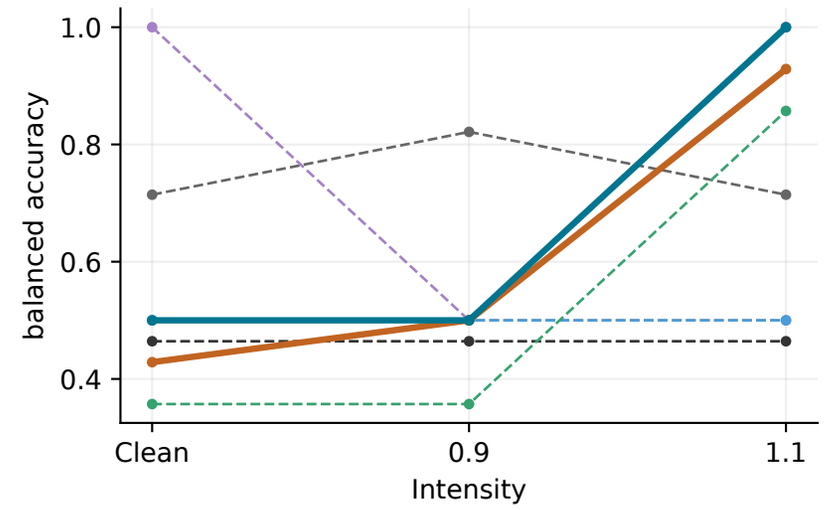

Reused OpenNeuro: 15 patients (1 AD, 14 CN) | ROC and precision–recall
AD sensitivity
False-positive rate
AD precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0
Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble
Small fixed case-control samples; precision reflects sampled prevalence. All scores and uncertainty appear in analysis.json.

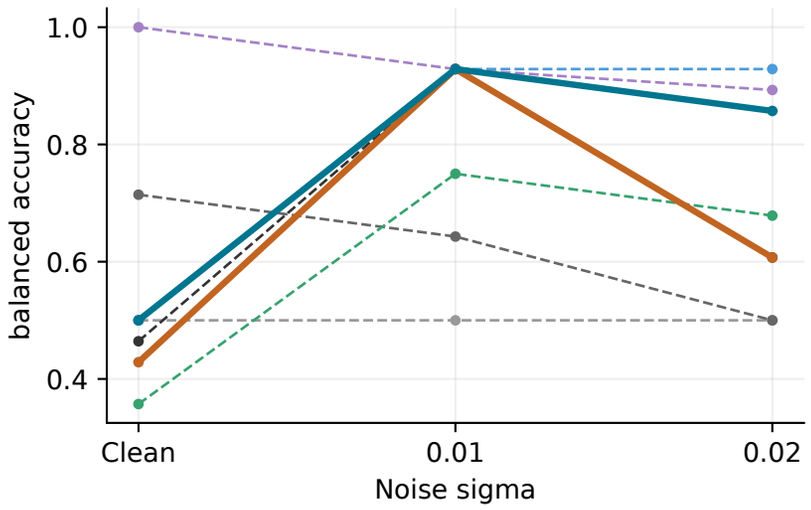


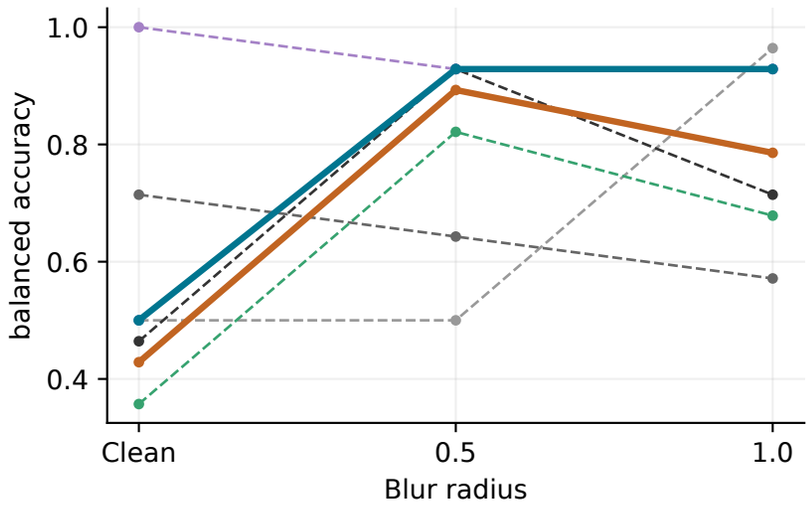
Five fixed probability bins are descriptive; sparse bins and case-control prevalence do not establish clinical-risk calibration.

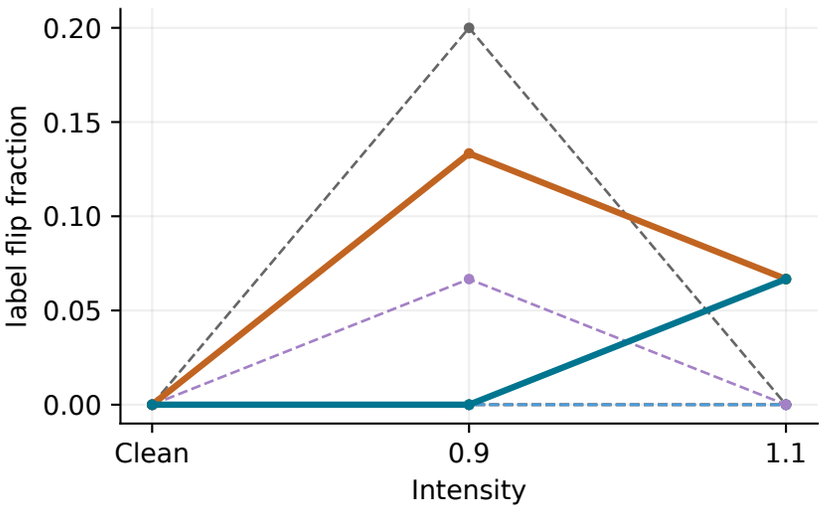

Reused OpenNeuro: 15 patients (1 AD, 14 CN) | confidence
Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble
Provider CN
Provider AD
Patient count
Patient AD score
Ten fixed histogram bins show patient scores by provider diagnosis; case-control score distributions do not establish clinical-risk calibration.

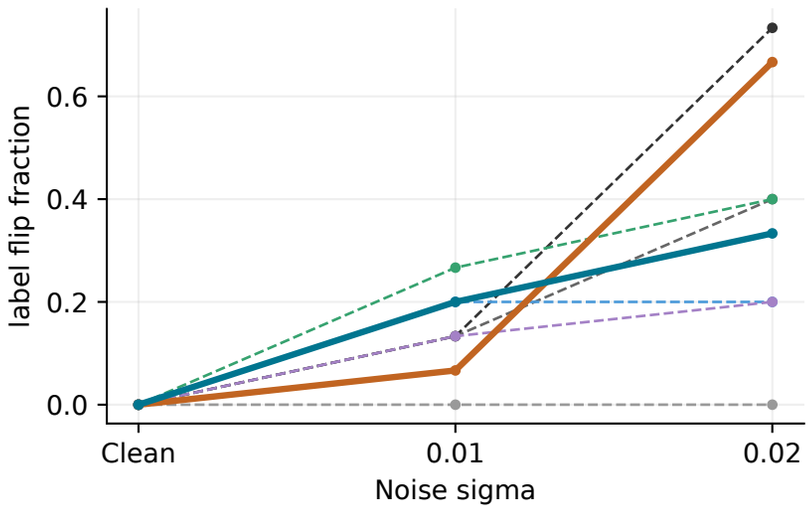
Reused OpenNeuro: 15 patients (1 AD, 14 CN) | all 7 fixed pixel-stress conditions

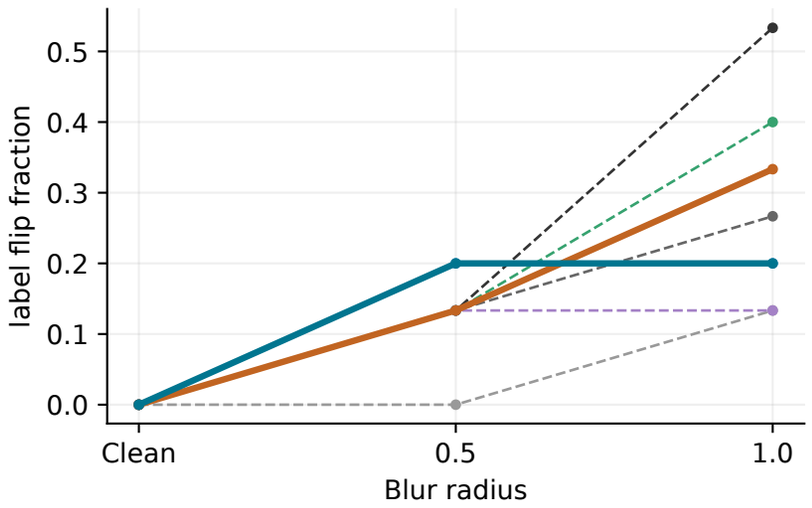


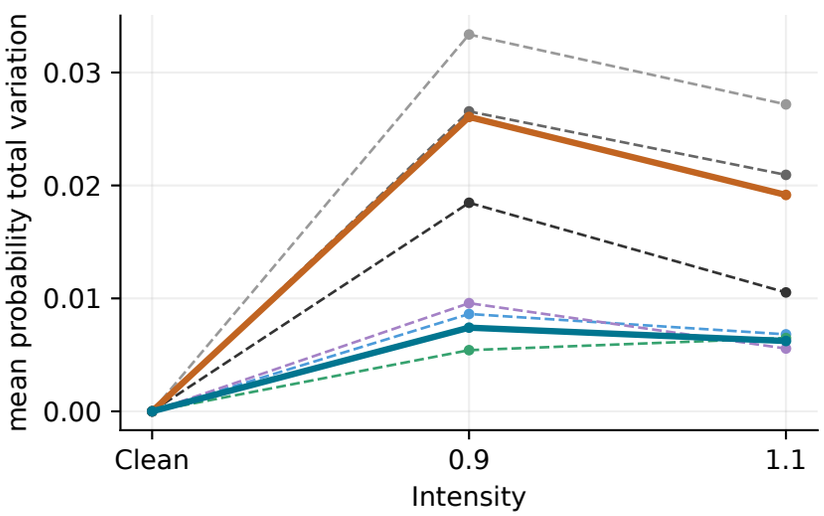


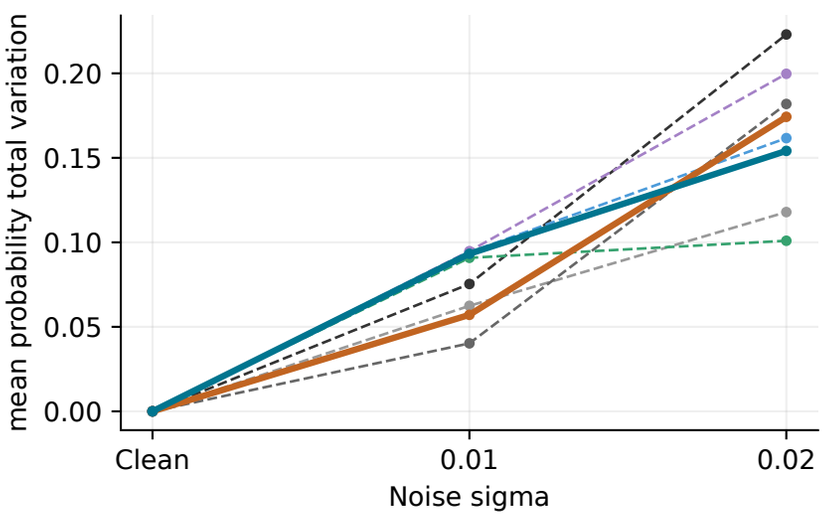


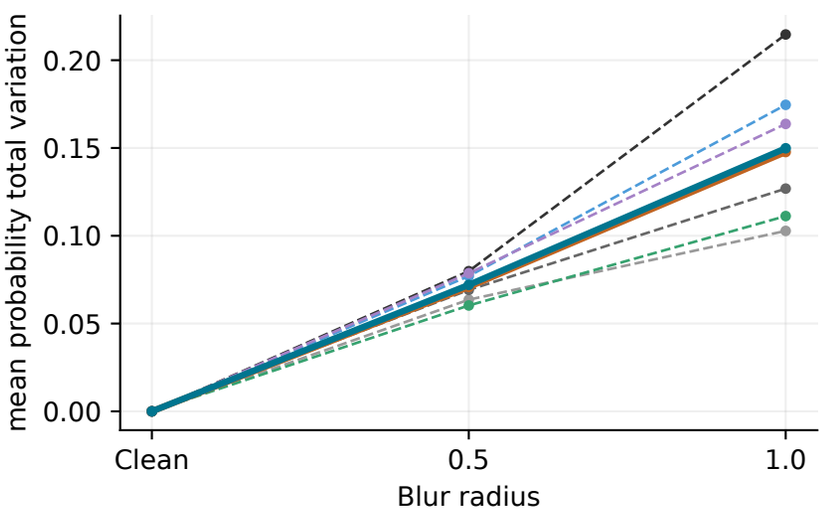












Old clinical c2
Old clinical c4
Old clinical c6
New seed 20261021
New seed 20261022
New seed 20261023
Old fixed ensemble
New fixed ensemble

Same patient, perturbation and clean reference for every model. These synthetic 2D controls do not simulate MRI acquisition physics.